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Business Model Elements for Sustainable Water Treatment Systems in Resource-Constrained Areas

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Abstract

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Emelie Axelsson Stenroos & Paul Kreis

While Sub-Saharan Africa has seen ongoing infrastructure expansion, a persistent functionality gap leaves roughly one-third of rural water systems non-functional or strongly underperforming at any given time. Community-Based Management has proven fragile in practice. While the Service Delivery Approach is an advancement, it operates more as a diagnostic framework rather than a design blueprint. Business model theory has not yet been thoroughly applied to essential services in resource-constrained areas, leaving a gap in how sustainable operations should be organized.

This thesis addresses the question: *What elements need to be considered for a business model of an off-grid water treatment system to be sustainable in a resource-constrained area?* An inductive, single-case study of a Swedish startup's pre-commercial Off-Grid Water Treatment System is conducted, drawing on stakeholder interviews and document analysis in Kenya and Zambia. Sustainability is defined as reliable drinking water delivery over ten years without any breakdown exceeding 48 hours.

Three interdependent themes emerge from the findings. First, trust, local ownership, and accountability are prerequisites for governance. Support from traditional leadership and community participation in planning proved more important than formal contracts. Second, pay-per-use tariffs alone are insufficient given seasonal agricultural incomes. Cross-subsidization through productive enterprises that use water commercially emerged as a more viable financial architecture. Third, technical failure is primarily a people and incentives problem. The brain drain of newly trained operators leaving for opening urban employment opportunities and a lack of maintenance culture are predictable structural patterns that must be accounted for in business model design.

By combining these governance, financial and technical dimensions, the study proposes business model elements for organizing essential services in resource constrained-areas, where standard market assumptions cannot be assumed.

Keywords: Business model, business model elements, resource-constrained area, water treatment system, sustainability

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Popular Science Abstract

Around one in three water systems built across Sub-Saharan Africa are non-functional or severely underperforming at any given time. The technology is rarely the cause for this.

The most common governance model, of volunteer community committees responsible for collecting fees and organising repairs, has well-documented weaknesses. Committees lose members, fee income stops when harvests fail and trained operators leave for better opportunities elsewhere. When donor funding ends without a transition plan, communities often wait for the next external project to fix it.

This thesis asks what a business model for an off-grid water treatment system needs to look like to remain functional for ten years or more. The empirical case is a Swedish startup's pre-commercial off-grid water treatment system, with interviews conducted among practitioners working in or with Kenya and Zambia.

Three themes emerged. The first is governance. Systems are more likely to be maintained when traditional leadership promotes the project, when community members have participated in design decisions and have developed trust and when specific individuals carry clearly assigned responsibilities. Ownership appears to be built through participation in the process, not through formal assignment alone.

The second concerns finance. Pay-per-use tariffs are unreliable where income is seasonal and irregular. The most viable alternative identified is cross-subsidisation through a productive enterprise that uses water commercially and whose revenue offsets what households cannot consistently pay.

The third concerns technical sustainability. One practitioner noted that many Kenyan languages have no direct word for maintenance, therefore it cannot be assumed that preventive maintenance will be executed without sufficient incentives. Operators trained to run systems can leave for urban employment once their skills are marketable. Distributing knowledge across multiple people and where possible, compensating operators adequately financially reduces that risk.

The pattern across all three themes is consistent. These are organisational problems, not technical ones. Where the business model accounts for them from the outset the chances of the technology working for a long time are increased.

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List of Abbreviations

CBM	Community-Based Management
SDA	Service Delivery Approach
SBM	Sustainable Business model
BMI	Business Model Innovation
FI	Frugal Innovation
BoP	Base of the Pyramid
OWTS	Off-Grid Water Treatment System
WEIHN	Water-Energy-Infrastructure-Human Nexus
WASH	Water Sanitation and Hygiene
NGO	Non-Governmental Organization
DRA	Demand Responsive Approach
OECD	Organization for Economic Co-operation and Development
TSU	Technical Support Unit
EU	European Union
LPM	Local Project Manager
COMP	Company
BIA	Business Incubator and Accelerator
GOV	Governmental Organization
SSS	Swedish Support System
ENT	Entrepreneur

1. Introduction

Access to safe drinking water is one of the most fundamental factors of human health, well-being and development. However, billions of people around the world do not have reliable access to it. In 2022, around 2,2 billion people lacked access to safe drinking water services. This means clean water available when needed and water access close to home. Unsafe drinking water contributes to the spread of diseases which is causing hundreds of thousands of deaths every year (World Health Organization, 2023). Better water access leads to reduced time spent collecting water, which could improve economic productivity and school attendance. In 2010, the United Nations declared access to safe and affordable drinking water a human right. Building on this commitment, the Sustainable Development Goal 6 aims to ensure universal and equitable access to safe drinking water by 2030 (World Health Organization, 2023).

Despite these efforts, progress is slow and current investment levels are still below what is needed to achieve universal coverage in many parts of the world (UNICEF, n.d.). In Sub-Saharan Africa, decades of investment have expanded infrastructure and contributed to progress toward Sustainable Development Goal 6 (Hutchings et al., 2015; Lockwood & Smits, 2011). Yet this progress hides a functionality gap as improvements in coverage do not automatically translate into sustainable water services. At any given time, a large share of installed systems are non-functional or severely underperforming. Those are not isolated incidents, they are more of a well documented pattern (Carter & Ross, 2016; Harvey & Reed, 2007). The binary distinction between working and non-working systems further hides the reality. True functionality requires reliability over time, not just installation. Times of partial breakdown and slow repair are common and the real measure of success is sustained service delivery (Carter & Ross, 2016; Foster et al., 2025). The implication is clear. The bottleneck is not a shortage of technology or initial financing. Systems are built and then they fail (Lockwood & Smits, 2011). The question is why and what can be done about it.

1.1 Background

The sector has evolved over time. Early efforts were top-down projects and centralized government delivery, which often failed (Moriarty et al., 2013). In response to high failure rates under this model, Community-Based Management (CBM) became the standard governance approach. CBM assigns ownership, fee collection and maintenance to local user committees, based on the logic that communities have the strongest incentive to keep their own water running (Cunningham et al., 2019; Harvey & Reed, 2007). Later CBM was complemented by the Service Delivery Approach (SDA). This framework argues for managing water as a continuous service over its full lifecycle, including professional support, performance monitoring and lifecycle financing, rather than as a one-time infrastructure project (Lockwood & Smits, 2011). These represent an evolution in thinking. A step was made from hardware provision toward service continuity and from external delivery toward local ownership (Moriarty et al., 2013).

Yet despite this evolution, failure rates remain high at around one third of them being broken, which shows the advances in thinking have not resolved the core problem. Literature suggests that the issue

is based on a lack of understanding of how to organize those ideas into a sustainable system rather than just not a lack of good technologies (Van Den Broek & Brown, 2015).

CBM has shown to be fragile in practice. It mixes fee collection with voluntary community labor in environments where communities often lack the financial capacity, technical skill and leadership to sustain both (Harvey & Reed, 2007; Kativhu et al., 2018). The consequences of collapsed cost-recovery, long repair response times, accountability gaps between central and local government and systems abandoned after handover are well-documented. These failures occur not because communities do not care, but because the organizational model does not allocate responsibility, resources and incentives in a usable way (Kativhu et al., 2018; Klug et al., 2017; Lockwood & Smits, 2011).

The SDA identifies that professional management, lifecycle financing and post-construction support is needed and functions as a diagnostic framework, not a design blueprint. It describes the destination without specifying how to get there organizationally (Lockwood & Smits, 2011; Moriarty et al., 2013). That design problem is what this thesis addresses.

1.2 Research Gap and Problematization

The problematization can be understood as an organizational design problem, one that has been proven to be difficult for the sector to address. The literature is strong on failure modes and weak on organizational design. It is known what goes wrong. We do not have an empirically grounded understanding of which organizational arrangements actually sustain water systems over time in resource-constrained areas. Further research is needed on which combination of actors, responsibilities, incentives and financial flows can solve this situation (Cunningham et al., 2019; Klug et al., 2017; Lockwood & Smits, 2011).

Addressing the question about organizational design requires a theoretical framework for how responsibilities, incentives and financial flows are structured across actors. Business model theory is that framework, but its application to essential public services in resource-constrained contexts has been limited (Chabowski et al., 2025; Onsongo et al., 2023). Sustainable business model thinking, which incorporates social and environmental value alongside economic returns, is relevant here as water service delivery cannot be measured by profit alone (Bocken et al., 2014).

However both Business Model Innovation (BMI) and Sustainable Business Model (SBM) research are mostly developed for and validated in stable, competitive market environments. These assume enforceable contracts, market-clearing prices, profit incentives and functioning institutions. But these assumptions do not reliably exist in resource-constrained areas in Sub-Saharan Africa (Massa et al., 2017; Onsongo et al., 2023). Frugal Innovation (FI) and Base of the Pyramid (BoP) research add some nuance on adapting solutions to resource-constrained environments. Yet, they tend to focus on product design and affordability rather than the organizational architecture that sustains those products over time (Onsongo et al., 2023; Rosca et al., 2017). No existing framework addresses resource-constrained context, essential service logic and organizational sustainability at the same time.

The result is that the water sector and business model research have talked past each other. Water literature knows what is organizationally needed but not how to design it (Onsongo et al., 2023).

Business model theory has the design tools but has not applied them to essential services in resource-constrained contexts. This is the gap this thesis addresses.

1.3 The Case Context

This thesis examines the research gap through a concrete case. One solution for an Off-Grid Water Treatment System (OWTS) is developed by a Swedish startup, named in this report as Water Tech. Some of its features directly address several recurring organizational failure drivers like spare part dependency, energy access and need for specialized maintenance staff (Water Tech, 2026).

The Water-Energy-Infrastructure-Human Nexus (WEIHN) framework captures why technical promise alone is insufficient. A system succeeds only when energy, infrastructure and human factors are aligned (Huang et al., 2023). The OWTS addresses the first two. The business model addresses the third.

The OWTS prototype is under construction and Water Tech has not committed to a business model for the Sub-Saharan African market. This is analytically significant. It allows exploration of organizational design without the path-dependency of an already-operating model. Who owns it, maintains it, finances it and bears accountability when it fails remains open (Water Tech, 2026). This openness provides a useful context for exploring how a sustainable business model could operate in practice.

1.4 Aim of the Study and Research Question

The aim of this thesis is to identify how the sustainability of an off-grid water treatment system in resource-constrained areas can be improved using a business model approach, looking at Water Tech and the OWTS as the empirical case. A community is seen in this thesis as a group of people that uses the same water source for their daily needs. Sustainability is defined in this study as reliable delivery of drinking water for at least ten years without breakdowns exceeding two days, drawing on the functionality criteria discussed in Carter & Ross (2016) and Lockwood & Smits (2011). This grounds the concept in service continuity rather than the binary functional or non-functional measure that dominates existing research.

Research question: *What elements need to be considered for a business model of an off-grid water treatment system to be sustainable in a resource-constrained area?*

An inductive, grounded approach is adopted. Theory is built from practice rather than deduced from frameworks developed for incompatible settings (Bell et al., 2019).

1.5 Delimitations

The study focuses on organizational and financial design during and post-deployment, excluding manufacturing and technology development. While no statistical generalizability is claimed, theoretical transferability is offered through rich description for comparable resource-constrained contexts. The geographic focus is on Sub-Saharan Africa, primarily Kenya and Zambia. The OWTS

was in the pre-commercial deployment stage at the time of research and findings reflect practitioner knowledge rather than post-deployment evaluation.

Some sources cited in this thesis come from literature with Water, Sanitation and Hygiene (WASH) sector evidence and frameworks. In these cases, the most relevant insights related to water have been extracted and applied to the focus of this study, since sanitation and hygiene fall outside of the scope.

The thesis is structured in the following way: Chapter 2 reviews the literature on business models, sustainability and the water context. Chapter 3 presents the research methodology. Chapter 4 reports findings from stakeholder interviews and document analysis. Chapters 5 and 6 discuss and conclude.

2. Literature Review

This section outlines how the existing literature relates to the research question of this thesis. Figure 1 shows how each subsection relates to the question and how it contributed to the formation of it.

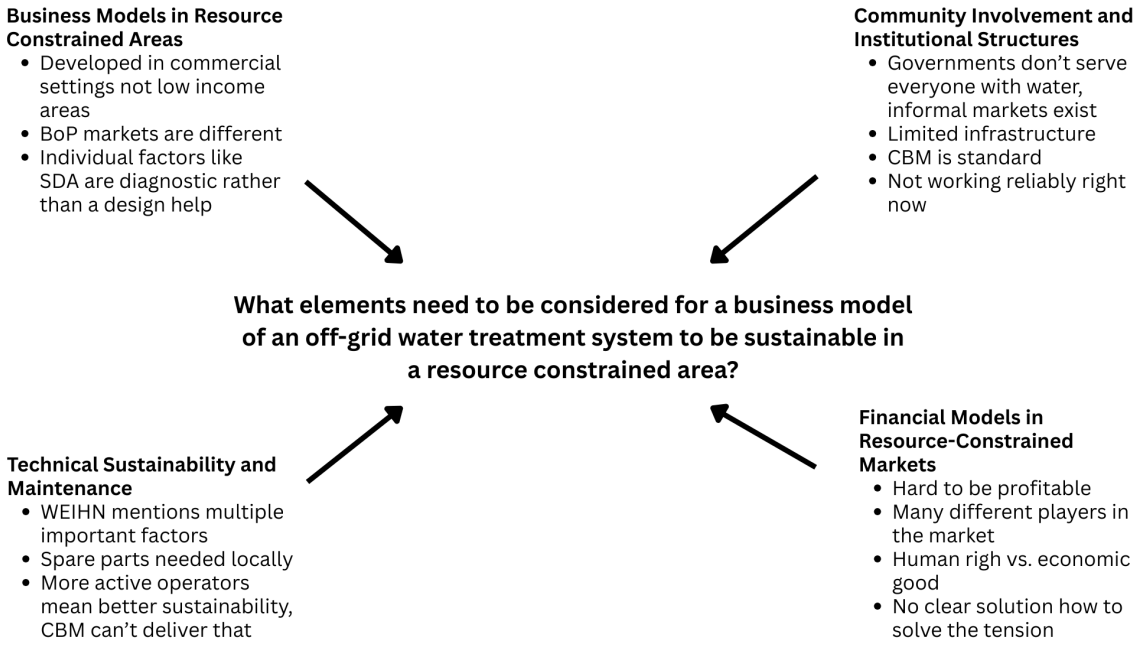


Figure 1: *Relation of existing Literature to the Research Question (Foss & Saebi, 2017; Gc et al., 2021; Huang et al., 2023; Lockwood & Smits, 2011; Onsongo et al., 2023; Van Den Broek & Brown, 2015)*

2.1 Business Models and Sustainability in Resource-Constrained Water Systems

To understand what makes a business model sustainable in resource-constrained areas, it is necessary to first look at how business models are described and how sustainability has been understood in the literature. Business models have traditionally been viewed as a way to organize commercial value creation, but more recent research highlights the limits of applying these ideas to resource-constrained areas. In these contexts, market conditions, institutions and social dynamics often work very differently from those in developed economies (Bocken et al., 2014; Onsongo et al., 2023). This section therefore starts with business model theory, then explores how sustainability frameworks apply in resource-constrained areas. The section ends with more information of the water sector to highlight the gap this study aims to address.

2.1.1 Business Model Theory

Early definitions have described business models to be how an organization functions and achieves its goals (Massa et al., 2017). More recent research describes business models to be more complex and highlights the importance of viewing it as a system of interdependent activities involving multiple actors and stakeholders (Zott & Amit, 2010). Business models can be understood in different ways but

are often seen as frameworks that organize how operations are conducted (Bocken et al., 2014). A business model can also act as a base to guide decision-making and provide a description of how value is created and captured (Massa et al., 2017).

In the context of a business model, Zott and Amit (2010) describe there are three design elements included. Those are content, structure and governance, which define how a firm together with its partners and customers creates value. With the elements grouped into themes, they are described in further detail in the NICE design. As shown in Figure 2, the NICE framework highlights four important dimensions: novelty, lock-in, complementarities and efficiency. These together show how new activities are introduced, how relationships are sustained, how activities are dependent on each other and how finances are managed in the system (Zott & Amit, 2010).

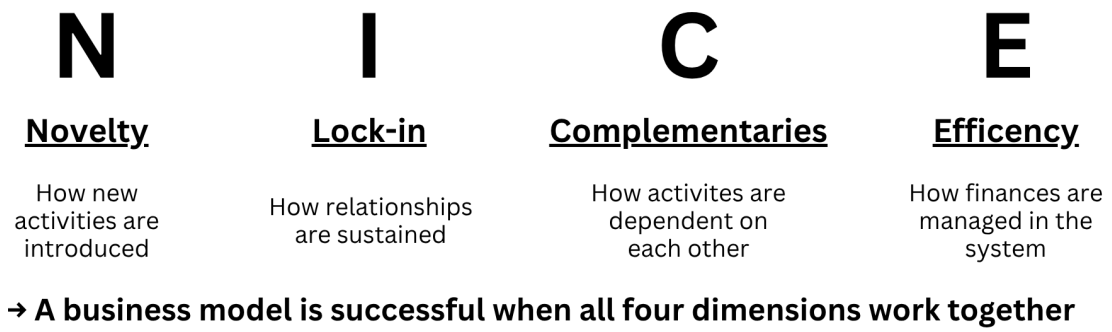


Figure 2: *NICE Framework adapted from Zott & Amit (2010)*

Another dimension included in a business model is proposed to be the Magic triangle (see Figure 3 on page 7), which is often used to answer four questions: who the customer is, what value is offered, how that value is delivered and why the model generates financial returns (Gassmann & Frankenberger, 2025).

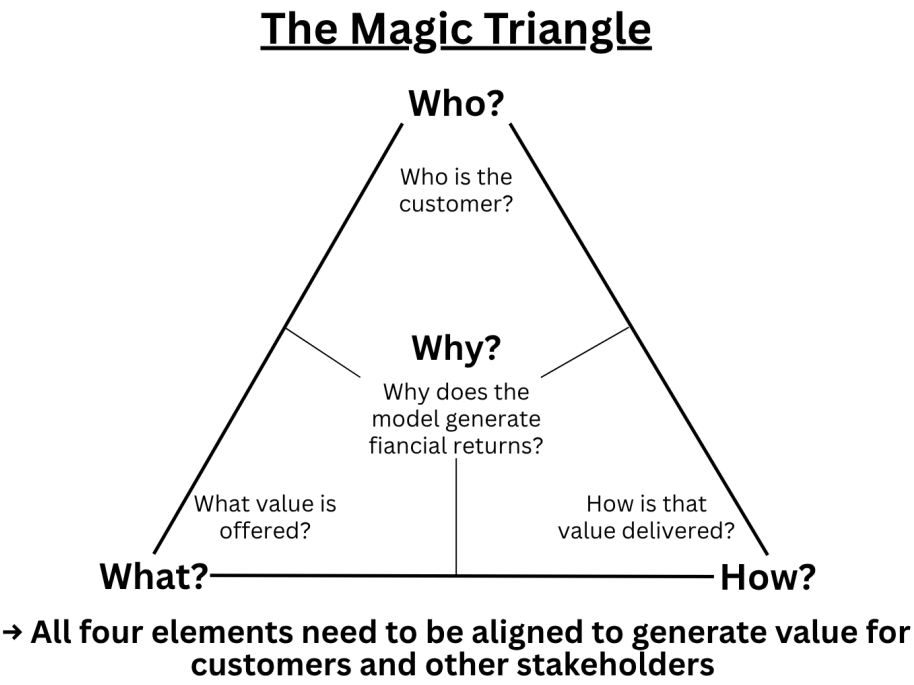


Figure 3: *The Magic Triangle adapted from Gassmann & Frankenberger (2025)*

Together, these frameworks represent some of the main ways tasks are governed and captured in a business model design in previous literature. These perspectives reflect how business models have been described by researchers. However, one important aspect that is being pointed to is how many models are still restricted to economic value and profit maximization with a focus on commercial markets (Foss & Saebi, 2017).

2.1.2 Sustainability

SBMs are characterized with complex tasks and are often associated in literature as the Triple Bottom Line (TBL). The TBL framework is discussed to expand the traditional business models as the focus includes not only the economic perspective, but also the social and environmental (Schoormann et al., 2022). Research is approached with the view that a business should integrate responsible resource use and long-term value over short term gains (Toledano & Horie, 2025). However, to include such balance in low-income contexts can be challenging as companies become limited to customers' ability to pay and to other factors that might exist in resource constrained areas (Gedamu et al., 2025). This could potentially weaken the ability to create financial sustainability (Bocken et al., 2014). What previous literature often highlights is the tension between social goals and financial viability, but rarely provides good examples of how this challenge can be managed in practice (Onsongo et al., 2023)

Literature on BoP markets and informal economics highlights that customers in these environments often have irregular incomes, which makes direct sales of services difficult. One other limitation

shown is how the value proposition may compete with free or low-cost alternatives (Molina-Maturano et al., 2020; Onsongo et al., 2023). Research highlights the BoP as an interesting market with big potential. Over half of the world's population belongs to this group, so even if people have low purchasing power, this customer group offers very high sales potential (Gassmann & Frankenberger, 2025; Massa et al., 2017). These markets are characterised by limited resources, weak or unreliable infrastructure, making new distributions and concepts necessary (Gassmann & Frankenberger, 2025). What the literature describes as institutional voids means there is a lack of well-functioning systems for markets, contracts, regulation and finance (Molina-Maturano et al., 2020; Onsongo et al., 2023). In the water sector, these gaps become visible when projects cannot be followed through or when systems lack formal support structures as it forces providers to operate under high uncertainty and increased costs (Onsongo et al., 2023).

Beyond sustainability, literature has increased its attention to the question of how business models are designed and adapted in resource constrained markets. An important aspect of that discussion is FI, which addresses innovation creation under conditions of limited resources. In literature, FI is often described as doing more with less, to simplify products and systems to reduce costs while still delivering value to low-income areas (Onsongo et al., 2023; Rosca et al., 2017). In a business model context, this means involvement for ongoing adoption and experimentation to fit local conditions while maintaining a constant value proposition (Onsongo et al., 2023). FI literature also highlights that innovation in resource constrained areas are dependent on interactions between several actors and institutions to support knowledge sharing and adaptation (Molina-Maturano et al., 2020).

FI often relies on a network of multiple stakeholders who share knowledge, distribute costs and work together to support adoption. This means FI rarely involves single actors when going into remote areas through local channels (Molina-Maturano et al., 2020; Onsongo et al., 2023; Rosca et al., 2017). Literature also highlights the role social embeddedness plays in sustaining innovation for a longer period of time. Integration of local materials, labor and community structures into the value chain both reduces costs and builds acceptance. According to researchers, acceptance is important as it creates a feeling of ownership, which is something traditional supply chains struggle to achieve in resource constrained contexts (Rosca et al., 2017). Inclusion of locals to participate in the project is described to be important as involvement strengthens engagement and commitment longer once external support leaves. Literature also points to a tension on how a business can provide high-quality solutions while still being affordable and at the same time ensure long-term financial sustainability. Companies must continuously look for a good balance between creating value and capturing it. The FI business model could act as a foundation for a BMI process (Onsongo et al., 2023).

BMI is discussed in literature to be more than product newness or diversified service offerings and is often associated with companies wanting to enter new markets. This means the path has to be formatted to go further than existing processes and products, as innovating the value proposition would most likely result in a product innovation (Bocken et al., 2014; Foss & Saebi, 2017; Massa et al., 2017). The literature suggests that new ideas as well as an organization's ability to learn, adapt and develop capabilities are important parts of successful innovation (Martins de Souza et al., 2024; Massa et al., 2017). In literature, it is highlighted that when two out of four elements in the Magic Triangle are changed, that is when a business model becomes innovative. Literature suggests that small changes, like recombining existing patterns can help create successful innovations (Gassmann

& Frankenberger, 2025). Much of the existing literature for BMI is based on competitive markets and does not give as much attention to public services. In those sectors, success includes reliability, fairness, public trust but also profitability (Massa et al., 2017; Onsongo et al., 2023). This results in limited research of how innovation and business models work together in resource constrained settings where commercial frameworks cannot be applied. In these environments, business model success is also connected to maintenance, long-term support and the ability for systems to continue working over time.

The water literature points to the lack of long-term support, asset management and institutional capacity as valid reasons why many systems fail over time. Research on rural water systems in Africa discusses that community-managed systems become non-functional which raises concerns about sustainability (Hutchings et al., 2015). About 25 % of rural water systems in developing countries are considered non-functional within two years of installation (Chowns, 2015). The result of this is described in literature as systems are left with long down time, leading to decreased quality and sometimes the system is even abandoned (Gedamu et al., 2025). Literature also points out that these business models struggle with scalability and long-term management as they often rely on informal structures and voluntary work (Moriarty et al., 2013).

2.1.3 The Water sector

While this study focuses on business models and long-term functionality for off-grid water systems, some of the literature used refers more broadly to water within the WASH sector (Gedamu et al., 2025; Hutchings et al., 2015). This reflects the practice of treating water, sanitation and hygiene as related areas linked to public health outcomes (Carter & Ross, 2016; Lockwood & Smits, 2011). At the same time, the sustainability challenges and technical requirements for water services are different in comparison to those related to sanitation. That makes the water topic a little bit different, which is why this study focuses specifically on the water topic (Lockwood & Smits, 2011).

Earlier research in the WASH sector has described limitations to be that many Non-Governmental Organizations (NGOs) and donors construct a system and then hand it over to the community without a long-term plan (Hutchings et al., 2015). Many of the gaps in resource-constrained areas with water treatment systems could be improved through the SDA. The SDA aims at improving long term functionality of water systems by sharing responsibility between the community and other actors, such as government and service providers. Compared to traditional community management, SDA focuses on professional maintenance and long-term support to make sure the system continues working over time (Chowns, 2015; Lockwood & Smits, 2011; Onsongo et al., 2023). This shift reflects a broader recognition in the literature that ensuring reliable and continuous service delivery is more important than simply providing access to infrastructure (Moriarty et al., 2013). However, even though the SDA represents an important shift in thinking, it mainly addresses how services should be delivered rather than how a viable and SBM should be structured around that delivery. Challenges around financial procedures, governance arrangements and technical capacity are each discussed separately in the literature, which is rarely integrated into a coherent framework (Chabowski et al., 2025; Gedamu et al., 2025; Lockwood & Smits, 2011). This leaves a gap in understanding how these dimensions work together in practice to support long-term sustainability in resource-constrained areas.

There is a gap in the literature regarding how SBMs can be designed and maintained in resource-constrained areas over time. Existing business model frameworks are many times developed in commercial and organizational contexts, focusing on competitiveness and market advantages which do not fully fit the conditions described in studies of water systems in Sub-Saharan Africa. Although the systemic view of business models is useful because value creation involves several interdependent actors, literature still provides limited understanding of how governance structures, technical capacity, financial mechanisms and contextual conditions are combined to support long-term functionality. This gap motivates the study, which takes an inductive approach to develop an empirically grounded understanding of the elements practitioners identify as important when designing business models for water treatment systems.

2.2 Community Involvement and Institutional Structures

In the water sector literature, there are suggestions of distinctions between governance and government. In research, government is referred to include formal rules and state authority, whereas governance is viewed in a broader way. Different actors, interests and relationships that influence and shape how water services are organized and delivered are all part of the picture of what governance is (Nelson et al., 2019). In resource constrained contexts, this becomes important as formal government structures are discussed to be weak or fragmented. In informal settings, governance is described to be dependent on a mix of institutional actions, social norms and informal structures, which is something traditional business models have not yet fully addressed.

2.2.1 Trust, Social Capital and Community Participation

Literature highlights trust and social capital as important factors for cooperation and long-term acceptance in resource-constrained areas and rural settings. For water projects, trust affects whether communities are willing to contribute to maintenance and continue supporting the system long-term. Wider discussions point out that lack of transparency or poor alignment with local social and cultural norms can lead to resistance and cause system failure. Literature also points to collaboration with local communities as an important way to build trust and increase acceptance (Harvey & Reed, 2007; Hutchings et al., 2015; Lockwood & Smits, 2011).

Rural communities often operate under community logic. This means projects are more likely to be trusted if they are led by a member of the community, which is a person belonging to the group. Outsiders, being people from outside the community, often lack established local networks. That makes it difficult to implement a system in these areas, but is essential even though it might be costly and time consuming to increase the level of trustworthiness (Fisher et al., 2017; Johanson & Vahlne, 2009). A firm can gain legitimacy and trust if they are supported by locals or recognized community organizations, as this signals acceptance in the community (Fisher et al., 2017). Research discusses that locals should be included in the planning phase to create a sense of ownership of the system. This is important for motivating people to pay and maintain the water system. Involvement also helps companies to deliver a product locals are in need of, ensuring the system is socially and culturally appropriate (Gedamu et al., 2025; Molina-Maturano et al., 2020; Onsongo et al., 2023).

The consequences of excluding communities from decision making could potentially lead to abandonment, if a system is left in villages without institutional capacity and a willingness to manage it. A lack of participation can also create a feeling of detachment. If any other actor, like an NGO or the government owns the system, people in the communities are hesitant to invest or maintain it (Gedamu et al., 2025; Hutchings et al., 2015). By excluding locals, they are not given the chance to learn about the system through training, making them very vulnerable and dependent on external actors to keep the system running (Gedamu et al., 2025).

Another aspect highlighted in literature is legitimacy. It is suggested to be an important part for understanding how water services are accepted by local members in communities. Research has identified three types: pragmatic legitimacy (performance based), moral legitimacy (based on shared values) and cognitive legitimacy (based on what is natural and taken for granted) (Suchman, 1995). Depending on which type of business model that is selected, it may rely on different types of legitimacy. For example, community-managed systems may build moral legitimacy, while professionally managed services may rely more on performance (Hutchings et al., 2015). In the context of water, literature suggests that legitimacy can be linked to participation. In newer research, it is shown that being included in the decision-making process is far more important than the final outcome. Additionally, trust is often formed from existing social structures as people in rural areas tend to trust traditional leaders more than formal committees (Harvey & Reed, 2007; Van Den Broek & Brown, 2015).

2.2.2 Institutional Frameworks and Responsibility Allocation

As highlighted earlier, the SDA is described as a common framework to use to understand how responsibility is organized. In the water sector, literature suggests it to be different between three levels: the national level (policy and standards), the intermediate level (planning and support, often through local government) and the local level (day-to-day operation) (Lockwood & Smits, 2011; Moriarty et al., 2013). The structure of the division helps explain where problems can come up when responsibility is distributed without the necessary resources or capacity. The idea behind the model is to make local community members take responsibility and thereby the ownership of the water system if they benefit from it (Harvey & Reed, 2007; Van Den Broek & Brown, 2015). This creates a shift from government led provisions, where centralized systems often failed and created lack of accountability and long term functionality, as seen in earlier research (Van Den Broek & Brown, 2015).

An important element that is described in literature is the Demand Responsive Approach (DRA), which includes expectations that locals contribute to initial investments. This can be done in different ways, for example, through labor, materials or money. The contribution is not only seen as a financial investment but also signals that people are willing to commit and is an attempt to strengthen the feeling of ownership. Within the existing community management model, responsibility is typically transferred after installation to a water user committee or a water point committee, which manages the system, collects fees and oversees repairs. (Lockwood & Smits, 2011; Van Den Broek & Brown, 2015). In research, it is highlighted that the success of DRA depends on formal and informal relationships. The relationships help minimize that people use the system without paying. It also encourages participation and supportment of developments and adaptability to local rules (Van Den

Broek & Brown, 2015). Previous literature also points out that communities can manage shared resources successfully under certain conditions, such as clear rules, monitoring and implementation (Dhoba, 2020). At the same time, studies point out that community based management has many times before been used as a one-size-fits-all solution without adapting to local context (Harvey & Reed, 2007).

While the literature discusses these formal roles within a community, research also highlights that these committees often face significant operational challenges. Committee members are usually unpaid volunteers, which can reduce motivation and engagement over time. Research also points to limited technical and administrative capacity, such as difficulties with maintenance and bookkeeping. Social relationships within the community can also make it difficult for committees to enforce rules or collect payments (Chowns, 2015; Harvey & Reed, 2007; Van Den Broek & Brown, 2015).

In a more formal setting, governance of a system often includes structured contracts and includes the agreement that governments are the provider to deliver water services. A common model that addresses this structure, is described in literature to be the Delegated Management Model. The aim with the model is to reduce risks, like poor service performance or lack of accountability (Gilson et al., 2025; Lockwood & Smits, 2011). However, the informal water markets are different as they tend to rely on rules that are unwritten but formed by trust, norms and interactions with community members. Informal rules are often described to be decentralized and based on actions in the everyday life, often called rules-in-use, which may differ from government regulations (Van Den Broek & Brown, 2015; Wutich et al., 2016)

Literature also discusses local government to be important even though their capacity may be limited. In many Sub-Saharan African contexts, rural district councils are formally responsible for water infrastructure and are expected to provide technical and financial support when needed. They are also supposed to consider part of their budget for water supply. However, this does not always happen in practice as researchers highlight that these targets are often not met, which shows a broader issue where formal responsibilities exist theoretically but are not fully implemented in reality (Kativhu et al., 2018). This shows that limitations need to be considered when designing a business model.

Another aspect that existing information about business models rarely includes is the role of external partners like NGOs. The literature on community-based management shows that many water systems rely on these actors for funding, technical support and training. The way these partnerships are structured matters as some approaches focus on building local capacity, while others rely more on external expertise (Cunningham et al., 2019; Kativhu et al., 2018). This becomes important for long-term sustainability, as systems that are dependent on external support can struggle once that support ends.

The shift toward service delivery identifies the need for ongoing support and maintenance. However, the gap between formal responsibilities and local capacity suggests that the sustainability for an off-grid water system depends less on who is officially responsible and more on the business model that is designed to manage resource-constraints in reality.

2.2.3 Risk, Incentives and Regulatory Enforcement

Risk is an important factor when talking about water governance. As literature describes it, the roles of different actors also come with different risks. For example, investors may carry a higher financial risk, whereas users in the communities may be the ones to take the direct consequence if the systems fail (Klug et al., 2017; Obeta et al., 2023). As highlighted before, communities may have limited capacity to take care of the system if it fails. Therefore, the distribution of risks is something that needs to be taken into consideration in the business model design.

Another perspective shown in literature is how incentives shape behaviours. If there is no clear structure or any contract connected to the work, it may lead to weak supervision that could affect the outcome negatively. Research shows the aspect of people wanting to complete a task as fast as possible to get paid, without considering the quality of the work. When quality is downprioritized, it might lead to system failure faster than expected (Marson & Maggi, 2018). While the informal markets can support price competition, quality and better service, this could also lead to people being excluded when profit motives conflict with social needs (Chumo et al., 2025).

In the areas of Sub-Saharan Africa, literature pin points regulatory enforcements to be weak. Where regulation is strong, that may lead to higher water quality and stronger accountability. In cases the regulatory is weak, governance relies more on informal systems, social norms and local relationships. This affects the design of the business model, as it needs to be taken into consideration for it to work in practice (Chumo et al., 2025; Lockwood & Smits, 2011; Nelson et al., 2019; Youngstedt et al., 2016). The way risks are shared, incentives are balanced and rules are enforced are important factors that need to be considered when designing a business model for resource-constrained environments.

Literature discusses trust, participation, institutional frameworks and governance as important factors. However, there is still limited understanding of how these elements can be combined into a business model for water treatment systems in resource-constrained areas. Existing research addresses community involvement, distribution of responsibility and regulatory conditions as separate topics rather than as interconnected parts of a SBM. This gap motivates the need for an empirically grounded understanding of how these social and institutional dimensions interact in practice.

2.3 Financial Models in Resource-Constrained Markets

A repeated challenge that is highlighted in water sector literature is the gap between the cost to have a water system functioning and the financial resources available in resource-constrained communities. Researchers have approached this issue from several different angles. While some focus on analyzing how community-based cost recovery works in practice, others look at the life cycle cost and how they are financed (Hutchings et al., 2015; Kativhu et al., 2018; Lockwood & Smits, 2011). A third perspective that is presented in literature is how informal water markets operate (Chumo et al., 2025; Gilson et al., 2025). The three perspectives together help create a picture of how water treatment businesses operate. However, the literature leaves an unanswered question about how financial sustainability can be achieved over the full lifetime of a system.

2.3.1 Cost Recovery and the Limits of Community-Based Financing

In literature, CBM is highlighted to rely on user fees as the main way to cover costs for a water system. In theory, these fees should be enough to cover payments related to maintenance, spare parts and other operations deemed necessary for the system to operate without government support (Van Den Broek & Brown, 2015). However, information from literature has shown that collecting fees rarely is saved for later issues. Many studies point to high failure rates, which are often linked to financial issues. Previously, it has been shown that water committees often struggle to collect water fees to cover maintenance costs (Dhoba, 2020). This ultimately leads to long periods of downtime and sometimes abandonment. Others discuss a similar finding where lack of engagement once the system was installed was suggested to be one of many reasons why systems fail in Sub-Saharan Africa (Gedamu et al., 2025).

Researchers describe willingness to pay as being influenced by factors such as perceived service quality, trust and local economic conditions. In low-income areas, many households face financial limitations, as the ability to pay is frequently hindered by extreme poverty. Households earning less than 20 USD per month must prioritize immediate needs such as food and healthcare before paying for water services (Gedamu et al., 2025). Trust and accountability are other important factors for water systems to continue functioning because if people suspect mismanaged or misused money by water committees, people will stop paying (Chowns, 2015; Kativhu et al., 2018). The free alternative sources, from rivers or rainfall, also reduces a community's willingness to pay (Gedamu et al., 2025). People in agriculture communities are also very dependent on seasonal incomes, which makes monthly tariff models misaligned with economic cycles used (Chowns, 2015; Kativhu et al., 2018). Furthermore, connected to the DRA, which assumes that initial contributions creates a feeling of ownership and long term willingness to pay, research does not show any automatic relationship between the two. However, what literature discusses is how community members feel they have already paid for the system when contributing in the initial step. This makes people expect that maintenance should be taken care of by external agencies (Harvey & Reed, 2007; Van Den Broek & Brown, 2015).

The financial problem is further suggested to be connected to the human rights paradox (Van Den Broek & Brown, 2015). Literature highlights how water is treated as an economic good, requiring cost recovery, while it on the other hand, is seen as a basic human right making it difficult to push for mandatory payments. In theory, communities can deny those who do not pay, however, this approach is often unacceptable as people pay as much as they can with the money they have (Foster et al., 2025). These findings highlight a mismatch between the financial logic of the CBM and the financial reality in resource constrained communities. While this problem is well documented, the literature has not yet provided a clear alternative.

2.3.2 Lifecycle Cost Analysis and the Structure of Water Finance

As briefly touched on before, previous literature has approached financial sustainability through life cycle cost. This approach looks at the total costs required to keep a water system running over time, including operation, maintenance and support (Moriarty et al., 2013). While the life cycle cost helps

with clarification of what is needed financially, it does not explain how those funds should actually be raised.

In some literature, the Organization for Economic Co-operation and Development (OECD) framework is highlighted and identifies three main sources of finance: tariffs (user payments), taxes (public funding) and transfers (external aid). The three T's model is used to understand water financing and points out that no single source can work on its own (Lockwood & Smits, 2011). It is shown that user fees alone are often too low, public funding is limited and external aid can create dependency. What remains unclear is how these different sources should be combined in practice to support long-term financial sustainability, especially in different types of resource-constrained environments. This is one of the areas where more empirical insight is needed.

Literature describes a shift in how water systems are financed, from short-term infrastructure projects toward a SDA that focuses more on long-term functionality and financial planning, as shown in Figure 4. This includes moving away from full dependence on grants and instead combining different funding sources and revenue models to support more sustainable water systems over time (Lockwood & Smits, 2011; Onsongo et al., 2023).

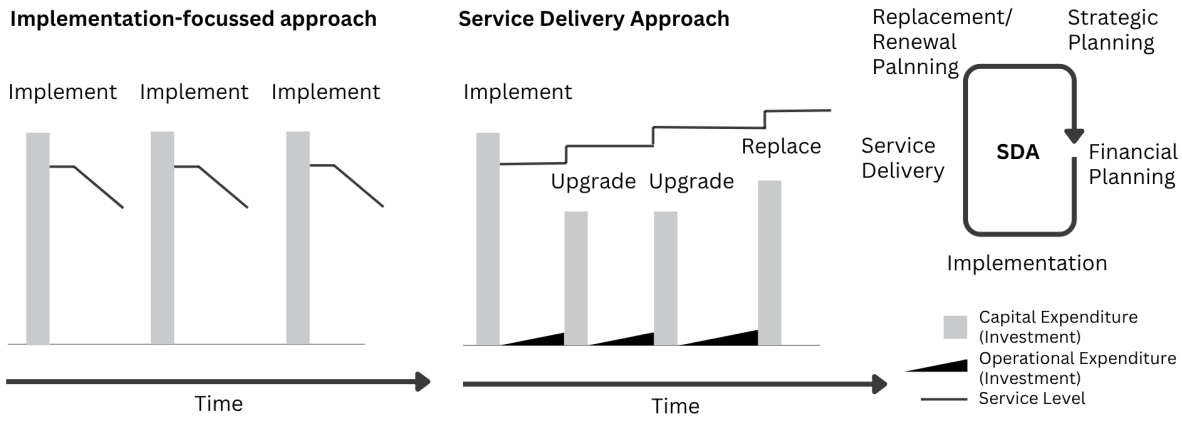


Figure 4: *Service Delivery Approach compared to previously common Implementation-focussed Approach, based on Lockwood & Smits (2011)*

External donor funding is often used to cover infrastructure investments, while user fees are expected to finance operation and maintenance costs. Central and local governments are expected to cover operating and minor maintenance costs through user fees. The DRA highlights that 10 % of initial capital costs are expected from communities to signal commitment and ownership (Lockwood & Smits, 2011; Van Den Broek & Brown, 2015). Research also highlights newer hybrid and frugal business models that combine different revenue sources, including income from other commercial activities, to improve long-term financial sustainability (Bocken et al., 2014; Onsongo et al., 2023).

Literature describes the transition from grant dependency toward financial self-sufficiency as a move toward more professionalized management. This can include paid and trained staff and more

professional work with management and financing. However, solutions like this are often easier to implement in communities with stronger financial conditions and more stable income sources. This creates dependency on external institutional support (Hutchings et al., 2015; Lockwood & Smits, 2011). Researchers highlight risks that are linked to strong dependence on donor funding. Lack of long term planning for maintenance once external support ends, together with rushed implementation processes creates a bad foundation for long term success. Research also suggests that heavy reliance on community management can reduce long-term responsibility from governments for maintaining public water systems (Gedamu et al., 2025; Kativhu et al., 2018). Overall, financial planning should cover the full lifecycle of water systems to include long-term maintenance and replacement costs that are many times neglected. Research also highlights the importance of pooled resources where multiple communities or water groups share financial resources together to help cover maintenance, repairs and other costs that may occur over time (Lockwood & Smits, 2011).

2.3.3 Informal Market Dynamics and Value Capture

Informal vendors many times act as an alternative system and provide water in areas where formal infrastructure is missing or unreliable. They have an important role in reaching last-mile users who are not served by centralized systems. Previous literature connects this to the invisible hand, where vendors respond to supply and demand without central coordination. In some cases, this leads to competition that improves price and service quality. The way these markets operate can be understood as a form of self-organising system (Chumo et al., 2025). At the same time, it can also result in uneven access and variable quality as these markets follow a different financial logic than formal systems. They often offer flexible, pay-as-you-go options, which suits well with the income patterns of low-income households (Gilson et al., 2025). This flexibility makes water more accessible, as people do not need to transport themselves long distances or contribute with large upfront payments. However, informal systems face their own challenges. Vendors usually have higher costs due to transport, storage and smaller sales volume and do not benefit from government funding. This makes water in the informal market often more expensive per litre than water from formal systems. Pricing is unpredictable and varies depending on distance, volume and delivery method (Gilson et al., 2025; Munro & Kweka, 2021).

Another aspect highlighted in literature is that systems allow for flexibility, but it also opens up opportunities for unfair practices. For example, vendors may charge full price even if they deliver less water. This may lead to restricting competition on the market, or avoid serving communities that are less profitable (Sarkar, 2020). This creates a tension between flexibility and fairness, where the same features that make the system work can also make it problematic.

The scale of the informal economy in Sub-Saharan Africa suggests that informal water markets are not a temporary gap-filling mechanism but a structurally embedded part of how water is accessed and distributed in these contexts. With informal activities, reaching over 85% of employment in the region, the majority of potential water customers operate outside formal economic structures (Gilson et al., 2025; Wutich et al., 2016). This has direct implications for business model design, as models built around formal payment systems, predictable revenue streams and regulated market structures may be fundamentally misaligned with the economic reality most communities experience. A SBM

for water treatment in these contexts must therefore account for the informal economy as a permanent structural condition rather than an exception to be overcome.

While individual financial challenges are well documented, existing research rarely explains how the elements of cost recovery, lifecycle financing and informal markets should be combined to reflect the economic realities communities face. This is important in resource-constrained contexts where no single financing source can be expected to be sufficient on its own and where formal business model logic does not align with the economic structures most communities operate within.

2.4 Technical Sustainability and Maintenance

While first-time access to water is growing more and more, the long-term functionality of these systems remains very low. Previous state-led or donor-funded approaches were common that focused mostly on building physical hardware quickly to meet coverage goals. However, research by scholars such as Moriarty et al. (2013) shows that focusing only on implementation is a main cause of system failure. This is because it treats water supply as a short project with an end date rather than a service that must last forever (Lockwood & Smits, 2011). This situation is made more difficult by economic water scarcity. This happens when water is available in nature, but people lack the money, skills or institutions to use it effectively (Lockwood & Smits, 2011).

2.4.1 Decentralized Water Treatment Systems and the WEIHN Framework

Water treatment systems are integrated technological infrastructures consisting of physical hardware, like wells, pumps and distribution networks and specific biochemical processes, primarily filtration and disinfection, designed to produce safely drinkable water. These systems are categorized into centralized models, which manage large-scale treatment and distribution from a single central unit and decentralized or point-of-use systems, which utilize localized, on-site processes to meet water needs in remote or resource-constrained areas (Huang et al., 2023).

To understand how decentralized systems should be used, the WEIHN framework can be looked at (Huang et al., 2023). The WEIHN framework recognizes that a system succeeds only when there is a strong link between energy sources, the physical infrastructure and the people who operate it. Huang (2023) argues that most research focuses only on the flow of resources and fails to explain how infrastructure and human factors interact. In resource-constrained areas, there is high potential for systems that use low-input and consumable-free technologies. However, these off-grid systems introduce new risks. For example, decentralized infrastructure can be more vulnerable to local environmental changes compared to simple mechanical pumps. A SBM must therefore address all parts of the WEIHN nexus to ensure reliability (Huang et al., 2023).

2.4.2 Maintenance Success Indicators

A debate in water research is how to measure success. Carter and Ross (2016) have criticized the use of binary indicators that simply say a system is either functional or non-functional. These simple measures are called a blunt instrument because they do not show the reality of repair cycles. A survey might record a system as failed just because it is temporarily being repaired on that day. Carter and

Ross (2016) argue that true sustainability requires looking at downtime durations and the history of repairs, so over what period of time a sustained service delivery happened. This also requires a focus being able to cover any upcoming Capital Maintenance Expenditure. This can be defined as the high, irregular costs of renewing major assets and doing large-scale repairs (Lockwood & Smits, 2011).

For repairs that are too complex for a local community, the literature identifies Technical Support Units (TSUs), as an option. Lockwood and Smits (2011) describe TSUs as external, professional groups that often work at a regional government level to provide technical help and backstopping to local operators. In Uganda, for instance, TSUs were created to help local governments manage major technical failures that were beyond their local skills (Lockwood & Smits, 2011).

2.4.3 The Tensions in Community-Based Management

The human part of the WEIHN framework has to provide the effort needed to keep systems running (Huang et al., 2023). However, the traditional model of CBM has been criticized as an insufficient way to supply rural areas. Based on research in mid-west Uganda, Van Den Broek and Brown (2015) argue that this model is a blueprint for breakdown. This is because it tries to combine profit-based fee collection with voluntary community action. This model often relies on grassroots romanticism, which is the false assumption that rural communities are always harmonious and will volunteer their time forever (Van Den Broek & Brown, 2015).

Van Den Broek and Brown (2015) show that when a voluntary committee has to collect fees, it can destroy the potential of collective action and introduce social tensions. Their research in Masindi and Kiryandongo districts found that people are often unwilling to chase their neighbors for money, which leads to a reduced interest in volunteering and the system falling apart. As a result, research is looking toward the professionalization of operators (Moriarty et al., 2013). Evidence from Nepal shows that the activity level of a trained operator is a very good way to predict how long a breakdown will last. Systems with active, paid professionals are fixed much faster than those managed by volunteers (Gc et al., 2021).

2.4.4 Logistics for Resilience of Spare Part Supply Chains

Even if the human part of the WEIHN framework is solved well, the supply chain for spare parts is an essential operational challenge that stays. Lockwood and Smits (2011) note that in rural areas, these chains are often very unreliable because the market for specific parts is too small to attract private businesses. Research suggests that sustainable operations depend on the national standardization of technology. Standardization means using the same technical designs across a whole country, which creates a large and predictable market that encourages local stores to keep parts in stock. This enables the sector to move away from temporary stocks kept by NGOs that disappear when a project ends to a more sustainable spare part market (Lockwood & Smits, 2011).

Sustainability is also improved when providers form associations and buy parts in bulk. This reduces the cost for each individual and gives the group more power to negotiate with sellers. For example, in Burkina Faso, the pooling of resources allowed communities to manage assets more efficiently (Lockwood & Smits, 2011). For a treatment system to stay functional, its business model must be part

of an enabling environment that coordinates government policies and private retail supply chains (Lockwood & Smits, 2011; Schiedek et al., 2021). This includes both construction and ongoing maintenance. As described earlier, the SDA was established from limitations in earlier research for the build and hand over process. In this earlier model, infrastructure was often provided by external actors, for example NGOs and then handed over to communities. The expectations were that locals would take responsibility for operation and maintenance (Moriarty et al., 2013). Furthermore, it is noted in other research that the SDA focuses on the initial construction phase, with the assumption that systems would continue to function without problem after handover (Lockwood & Smits, 2011).

As established throughout this chapter, many factors can lead to the breakdown of a water system in a resource constrained area. While the research explains these needs in theory, it does not show how to design a business model that actually provides them for the specific class of low-input, off-grid water treatment technology studied in this thesis. This thesis addresses that gap by looking for the specific elements needed to turn these technical possibilities into a permanent and reliable water service for areas with limited resources.

2.5 Synthesis and Gap

The literature reviewed in the previous sections provides several important insights about business models. There are many different elements that shape the sustainability of water systems in resource-constrained areas. Throughout the literature, researchers focus on specific aspects of the problem. The water sector literature provides more context-specific insights. It shows how governance structures, community-based management and service delivery approaches shape how water systems function in practice. It also highlights common financial challenges which includes the limits of cost recovery, the importance of lifecycle financing and the role of informal markets.

Addressing the question about organizational design requires a theoretical framework for how responsibilities, incentives and financial flows are structured across actors. Business model theory is that framework, yet it has rarely been applied to essential public services in resource-constrained contexts (Chabowski et al., 2025; Onsongo et al., 2023).

Business models describe how an organization creates, delivers and captures value, encompassing the full architecture of actors, activities and the underlying financial logic (Gassmann & Frankenberger, 2025). Sustainable Business Model (SBM) literature extends this to incorporate social and environmental value alongside economic returns. This is relevant to water service delivery, where success can't be measured by profit alone (Bocken et al., 2014).

However both BMI and SBM research are mostly developed for and validated in stable, competitive market environments. These assume enforceable contracts, market-clearing prices, profit incentives and functioning institutions. But these assumptions do not reliably exist in resource-constrained areas in Sub-Saharan Africa (Massa et al., 2017; Onsongo et al., 2023). FI and BoP research add some nuance on adapting solutions to resource-constrained environments. Yet, they tend to focus on product design and affordability rather than the organizational architecture that sustains those products over time (Onsongo et al., 2023; Rosca et al., 2017). For international business model adaptation, BoP markets present a separate challenge. They differ not only from developed markets but from each

other. This means they need simultaneous global and local alignment that standard business model frameworks do not address (Chabowski et al., 2025).

Finally, the contextual literature highlights what the reality looks like regarding institutional voids and weak infrastructure. It also discusses reliance on external actors and the importance of local adaptation. With the reviewed literature in mind, they lead to the gap that institutional, financial, technical and contextual factors are rarely analysed together. Most studies focus on one dimension at a time and existing frameworks do not provide a clear understanding of how these elements interact in practice. There is a lack of empirically grounded insight into how different elements come together in real-world settings and how people navigate the trade-offs between affordability, reliability and long-term viability. As a result, there is still limited knowledge of what actually makes a business model of an off-grid water treatment system sustainable over time.

This study addresses this gap by taking an inductive, grounded approach. Rather than applying a predefined framework, it explores how key elements emerge from practice and how they are combined to support SBM design in resource-constrained areas.

3. Methodology

Investigating which elements experienced professionals see as important for creating a SBM in resource-constrained areas requires methodological choices that can capture the specific issue at hand. The selected study approach is based on the idea that the elements needed for a SBM cannot be fully understood through measurements. They require a deeper exploration of understanding real life examples in a local context where water systems are operated. This chapter explains the methodological choices made for this study.

3.1 Case Introduction

The selected case for this study was a Swedish water technology startup and their OWTS. Water Tech was selected as the case study for several reasons. First, the OWTS system integrates established technologies in a novel design, promising significant advantages. Those advantages are minimal maintenance requirements, very low energy consumption, no consumable substances are needed, possible fast deployment of the unit in under four hours and ease of operation. These features are especially valuable in settings where technical expertise and resources are limited (Water Tech, 2026).

Second, Water Tech is currently constructing its first full-scale prototype, providing a unique opportunity to study the system before its commercial deployment (Water Tech, 2026). This early-stage focus avoids potential biases associated with existing business models or operational dependencies, which could be present in more mature companies. Water Tech's openness to exploring different business models further enhances the relevance of this case, as the company has not yet committed to a specific approach. Instead, it is actively seeking the most effective and sustainable model for both end-users and its own long-term success (Water Tech, 2026).

The core of this study is in analyzing the influences of different business model elements on the long-term functionality and sustainability of the OWTS system in resource-constrained environments.

Water Tech is focused on developing the OWTS system, a containerized water treatment solution. While the technology has been tested in laboratory conditions and is established in other water treatment settings, the full-scale prototype, as mentioned above, is currently under construction and has not yet been commercially deployed. This prototype is expected to be tested in late spring or early summer 2026 (Water Tech, 2026).

The OWTS system uses slow sand filtration and advanced oxidation, achieving efficient water treatment with a very low power consumption of approximately 0.5 Wh per liter. This energy efficiency allows the system to operate on solar power, supplemented by batteries, even in regions with low solar radiation such as Sweden. In Sub-Saharan Africa, where solar resources are more abundant, the system's energy requirements are easily met. Currently the concept does not include a water meter, so for any business models that include a pay-per-litre revenue stream the system has to be adapted for it in the future (Water Tech, 2026).

Maintenance demands are minimal. Weekly checks are limited to ensuring the water inlet remains unclogged, while the sand filter requires cleaning every few weeks, as soon as the flow rate of the water reduces. This is a simple process involving the cleaning of the top layer of sand. Advanced

oxidation components have longer replacement cycles of years, with tasks as straightforward as changing a lamp or plugging-in a new unit. The system is designed to be intuitive, requiring no specialized maintenance crew (Water Tech, 2026).

Through its containerized design, the OWTS system is portable and easy to deploy. Once placed near a water source (e.g., a lake or river), the system operates autonomously. Its design philosophy emphasizes ruggedness, simplicity and reliability, which should align well with the challenges of resource-constrained communities in countries like Kenya and Zambia (Water Tech, 2026).

Water Tech is currently applying for EU funding in collaboration with a local water utility company in Zambia and is in talks with partners in Kenya (Water Tech, 2026). While the findings of this study are relevant to any resource-constrained environment, the stakeholders approached in this study are mainly living in or operating in Sub-Saharan Africa. These regions are characterized by limited access to stable electricity grids, financial constraints and many peri-urban or rural settings. Understanding the economic realities of these communities, such as a very low daily income (excluding high income countries) of roughly $4.30 per day (The World Bank Group, 2026), is essential for designing a SBM.

This study will explore how different elements of a business model can facilitate the system's long-term functionality and thereby sustainability. Long term functionality of the unit is seen in this study as the system will deliver drinking water for at least ten years without major breakdowns that stop the production of drinking water for more than two days at a time. How and by whom the OWTS system is manufactured or delivered will not be looked at in this study.

3.2 Research Philosophy

This research was guided by an interpretivist epistemology and a constructionist ontology. Interpretivism focuses on understanding human behavior by exploring the subjective meanings people give to their actions (Bell et al., 2019). It was assumed the social world is different from the natural world because people are knowledgeable agents who can reflect on and explain their choices (Gioia et al., 2013).

From an ontological standpoint, constructionism suggests that categories like business models and their elements are not objective facts but are socially constructed and constantly revised through human interaction. This perspective emphasizes that functionality is not a fixed attribute of the technology itself, but rather a product of ongoing social processes and collaborations (Bell et al., 2019).

3.3 Research Design

This study was conducted as a qualitative single-case study. This design provided a systematic framework for investigating how different elements of a business model influences the sustainability of the OWTS within resource-constrained communities (Woodside & Wilson, 2003).

An explorative approach was chosen using grounded theory in line with the chosen research philosophy and how Strauss & Corbin (1998) view grounded theory. Specifically the Gioia

methodology was used to analyze the themes that emerged from the data. This allowed for identifying the unique nuances that should be considered for a business model in this specific context of a Swedish startup wanting to provide water treatment solutions to a resource constrained area in Sub-Saharan Africa (Bell et al., 2019). As the topic for this thesis had to be picked before the data was collected and analyzed, the researchers can not completely be considered a blank slate as they got in contact with related theory previously. However, the process described here in this chapter was followed throughout to reduce the chances of introducing biases and let the findings emerge from the data.

The specific mode of inquiry was theory generation for the given case. The empirical context of Sub-Saharan Africa was used to generate specific insights that go beyond general business model theories (Ketokivi & Choi, 2014). This satisfied the duality criterion of case research. The study remained situationally grounded in the specific challenges of the targeted regions while seeking theoretical insights that could apply to other resource-constrained areas (Ketokivi & Choi, 2014).

The study was idiographic in its focus, meaning the concern was with the unique, context-specific elements that allow business models to navigate the challenges of resource constrained environments (Bell et al., 2019).

3.4 Data Collection Methods

Data were collected using a triangulated approach, which involved gathering evidence from multiple sources to cross-check findings and enhance the credibility of the research. This method ensured that the limitations of a single data source were minimized by the strengths of others (Bell et al., 2019). The data sources included internal company documents from Water Tech, industry and NGO documents and semi-structured interviews with key stakeholders.

Document Analysis

A variety of suitable extant documents were analyzed that were not produced specifically for this research. These included Water Tech's internal documents, including a pitch deck, a product flysheet, a project application as well as the company website. From the NGO perspective a sustainability assessment and a project report were used. From the industry and financing perspective two strategy and investment documents as well as three projects reports were analyzed. These documents provided essential background information (Bell et al., 2019). A table with all data sources can be found in Appendix C.

Semi-Structured Interviews

The primary method of data collection was semi-structured interviews, which were designed as inquiry-based conversations (Castillo-Montoya, 2016). This format provided the flexibility to depart from the initial guide (see interview guide in Appendix A) to follow up on significant replies or pursue unexpected themes (Bell et al., 2019). To assist the interview questions, graphics for better understanding were shown to the interviewees for one more complex question (see Appendix B). As described before, participants were treated as knowledgeable agents capable of reflecting on their

actions and providing detailed explanations of their experiences with water system management (Gioia et al., 2013).

Interview Protocol Refinement

To ensure the reliability of the interviews, the Interview Protocol Refinement framework was followed as described by Castillo-Montoya (2016). This involved a four-phase process:

Alignment: An interview protocol matrix was developed to map each interview question directly to the research question, ensuring no information gaps existed.

Narrative Construction: Questions were worded in everyday language rather than academic jargon to remain accessible to participants.

Organization: The guide followed a logical flow, starting with non-threatening introductory questions to build rapport, followed by key questions related to business models and ending with reflective closing questions.

Feedback: The protocol was reviewed by an external person (a fellow masters candidate) to check for clarity, conciseness and the avoidance of leading questions.

Recording and Transcription

All interviews were audio-recorded using at least one digital device (phone, tablet and pc) to ensure the participants' answers were captured in their own terms and to allow the researchers to remain fully attentive during the conversation. Verbatim transcriptions were then produced for each interview to enable a detailed examination and analysis of the generated data (Bell et al., 2019). The interviews were conducted online or hybrid (one researcher on site, one online) using google meets, hostpoint and microsoft teams, depending on interviewee preference. In some cases the internet connection to the African interviewees was unstable throughout the whole interview. In those cases the questions and answers were repeated so every response was recorded in the end.

3.5 Sampling Strategy

The study used a purposive sampling strategy, selecting participants based on their strategic relevance to the research question rather than on a random basis. This approach ensured that those interviewed were knowledgeable agents capable of providing rich data on the functionality and sustainability of off-grid water systems within the context (Bell et al., 2019).

To ensure a comprehensive understanding of the operational dynamics of off-grid water treatment systems, a stakeholder map was developed to identify all relevant actors. This step was helpful, as the research addresses a complex social issue involving diverse stakeholders with varying interests and levels of influence. By systematically mapping these relationships, it ensured that the relevant perspectives were identified (Bell et al., 2019).

The initial stakeholder map was then validated with stakeholders on the company level. Their input confirmed the accuracy of the sampling frame and helped identify stakeholders, who might otherwise have been overlooked (Bell et al., 2019).

Sampling continued until theoretical saturation was achieved, the point at which new interviews no longer brought new insights or introduced new dimensions to the theoretical categories. Saturation was approached from interview ten onwards. The coding process was monitored to determine when additional data collection did not generate new insights anymore. This iterative approach ensured that the findings were both robust and grounded in the perspectives of the stakeholders (Bell et al., 2019).

Table 1 shows the 13 interviewees and the abbreviations used to reference them in the remaining of this work.

Anonymized ID	Descriptive Pseudonym
NGO-1	International NGO WASH Director
LPM-1	Local WASH Project Coordinator
COMP-2	Water Tech Chief Marketing Officer
COMP-1	Water Tech Founder and Chief Executive Officer
BIA-1	Social Enterprise Accelerator
GOV-1	Local Manager Technology Transfer and Commercialization
GOV-2	Local Enterprise Developer
GOV-3	Water Resource Authority Research Officer
SSS-1	Senior Sustainability Analyst
SSS-2	Director Export Finance
NGO-2	Senior Project Manager NGO
ENT-1	Local Social Entrepreneur (Energy/Water)
BIA-2	Local Director of Innovation and Entrepreneurship

Table 1: *Interviewees interviewed*

Note: See meaning of Abbreviations in List of Abbreviations on page X

3.6 Data analysis

The data analysis followed a systematic, multi-stage process designed to bring qualitative rigor to the interpretation of how different business model elements influence the systems sustainability (Gioia et al., 2013). The Gioia methodology was used, an inductive approach that allowed us to progress from raw stakeholder perspectives to high-level theoretical insights while maintaining a clear audit trail (Gioia et al., 2013). The analysis was treated as an iterative process, where data collection and interpretation occurred in parallel to refine emerging concepts and was conducted by both authors.

The first phase of analysis involved 1st-order analysis, which was performed through initial coding of the interview transcripts and documents using the software Taguette. The codes were kept closely to informant-centric terms, using the participants' own language to label segments of data without attempting to distill them into broader categories too early. This stage produced a comprehensive list (442 1st-order codes) of concepts that represented the experiences of the stakeholders (Gioia et al., 2013).

In the 2nd-order analysis, a transition to a more researcher-centric perspective to answer theoretically what is going on was used. The 1st-order concepts were exported from taguette to google sheets and analyzed to identify patterns, grouping them into broader, theoretically grounded themes (21 2nd-order codes). These themes were further distilled into three overarching aggregate dimensions, which served as the building blocks for the data structure. Figure 5 illustrates how the progress from raw data to the core theoretical dimensions of the research progressed (Gioia et al., 2013). An example of the coding process and the coding table can be seen in Appendix D.



Figure 5: *Example of process to progress form raw data to aggregated Themes*

Throughout the coding process, constant comparison was employed by both researchers, a technique where new segments of data were continually checked against existing codes and categories. This ensured a close fit between the evidence and the concepts, preventing cherry-picking quotes that only supported a specific narrative (Bell et al., 2019; Urquhart & Fernández, 2013).

In the finding section all quotes from the data are written in italic for a clear visual distinction between the researchers analysis and the original data. During the writing process, subsections were introduced within each theme to enhance the structure and readability of the text.

3.7 Research Quality

The authors recognize that the results are not intended for statistical generalization, as they are only based on one single case but rather to provide a credible and transferable theoretical account of how business model elements can improve the sustainability of water systems in resource constrained regions (Bell et al., 2019). As mentioned in this chapter the OWTS was not yet deployed while this research was conducted. Therefore all findings are based in the lived experiences of the interviewees and not in the analysis of a physical deployment of the system. To archive trustworthiness the following dimensions were considered.

Credibility

High credibility was ensured through triangulation, cross-checking data from semi-structured interviews against internal Water Tech documents, external NGO and industry reports. To improve the reliability of the research, the Interview Protocol Refinement framework was used to ensure every question was directly aligned with the research goals. Furthermore, by following the Gioia methodology, the development of a transparent data structure to display the findings as neutrally as possible.

Transferability

As the study is a single case study it can not be generalized or be easily transferred to a different setting. This is a general limitation of this research design. However a rich description and the full coding table are provided to allow other researchers to judge the applicability of the results to a similar environment.

Dependability

An auditing approach was maintained by keeping complete records of all research phases, including interview transcripts, coding lists and final analysis decisions (Bell et al., 2019).

Confirmability

Both researchers acted in good faith and tried to not allow personal values to influence the findings. While the case was conducted in collaboration with Water Tech, the process of conducting the research, collecting results and writing the thesis were not influenced by Water Tech, as the company was equally interested in understanding the context of this setting without any biases.

3.8 Ethics

It was adhered to four core ethical principles: avoidance of harm, informed consent, protection of privacy and prevention of deception (Bell et al., 2019). All participants were informed about the purpose of the research and their right to withdraw at any time. Informed consent for recording was obtained prior to interviews. Anonymity and confidentiality were maintained through the use of pseudonyms and secure data storage. All audio recordings will be deleted within three months of the completion of this thesis and only the transcripts will be stored. Reflexivity was practiced by explicitly disclosing the collaboration with Water Tech to participants and readers (Bell et al., 2019).

4. Results

This chapter explores the key findings from the research, looking at what is needed to create a SBM for off-grid water treatment systems in resource-constrained areas. Insights were gathered from a diverse group of stakeholders like local entrepreneurs, NGO representatives, government officials and technology providers working across or with partners from Sub-Saharan Africa. Their experiences highlighted three critical themes (shown in Figure 6) for long-term success that emerged from the second-order codes of the analysis. Building strong community involvement and institutional structures, designing financial models that fit the local context and ensuring technical systems remain resilient over time.

The research showed that sustainability depends on more than just technology or funding. Trust, adaptability and an understanding of local realities play a crucial role. It looks at topics like if a project can gain the support of local leaders, if it can address the unique needs and challenges of the community, if it will keep running when funding ends or equipment fails. These are questions that decide whether a water system will succeed or fail in the long run.

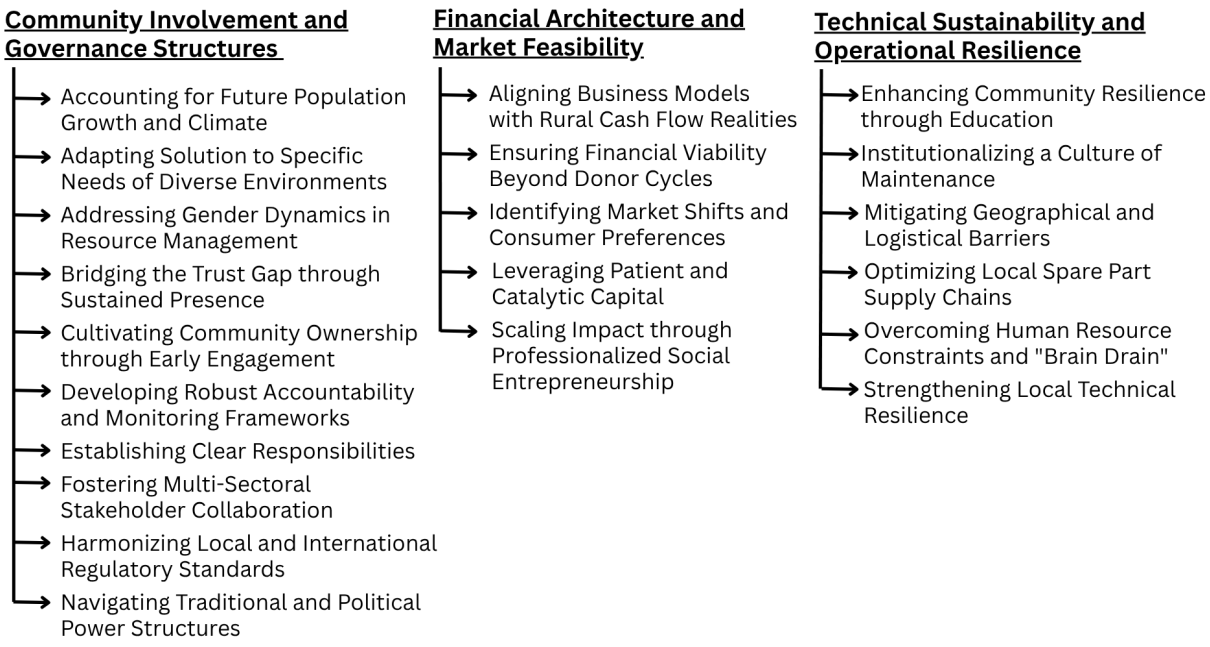


Figure 6: *Three themes for sustainability and the second-order codes they are based on*

4.1 Community Involvement and Governance Structures

Community involvement and institutional structures form the foundation for sustainable water interventions in resource-constrained areas. This section discusses, as shown in Figure 7 on page 29, success begins long before any technical system is installed, it requires earning trust through traditional leaders, navigating local norms that can override market logic and translating short-term acceptance into a genuine sense of community ownership. Without this groundwork, even

well-designed solutions risk being abandoned. In addition it also shows that formal governance frameworks often fall short when responsibility is delegated without adequate resources, creating accountability gaps that undermine long-term functionality.



Figure 7: *Community Involvement and Governance Structures Theme with Sub-Sections*

4.1.1 Trust-Building and Community Acceptance

Traditional leadership emerges from the data as the most important entry point for any water project as companies have to get a buy-in. This is made very clear by a local social entrepreneur who explained: "*before we start anything, we have to start with the local chief [...] Once he gives us his blessings, go ahead, green light [...] he gives a directive to the headmen, please go and support what these guys are doing [...] So that's a lot more important than even the government officials"* (ENT-1). They are the ones to give their blessing to actions in the community and can stop any activity in the area. The buy-in from the chief becomes a confirmation that the people understand the technology. There have been cases where sophisticated technologies have been introduced, however, they end up being abandoned. As one interviewee explained "*they just remain a black box, nobody knows what it is"* (LPM-1). Therefore, it is central for corporations to investigate the need in more detail, to tailor the solution to the problem. In many rural communities, chief meetings, funerals or church activities stop all economic activities, as described by this quote: *"These traditional leaders are very, very influential in the things that we do. Sometimes they will tell their farmers, if there's a funeral, for example, you find that business literally stops, because the community will go and support the bereaved family, so that one day there will be no production"* (ENT-1). These actions demonstrate that norms, rules and regulations outrank market logic and long-term strategy.

Interviewees highlighted that it is difficult to change people's minds. Even if there is a product that is clearly very profitable long term, they will not do it, as described here with some frustration:

> *That's a very difficult thing to change in our people. We have one product that is clearly very, very profitable, long term, and they could do more with it, but they won't do it. If you ask me, I don't even know how to explain it to them, because I've tried. The last two years I've tried, and it hasn't worked* (ENT-1).

This dynamic directly implements that new things that require change might be difficult to implement due to prioritizations in traditional behavior over economic growth. Therefore, when initiating a new water implementation program, it is highly recommended to have local partners involved that know the area and the people, as described by one interviewee: *"So when we implement these programs, we usually try to have one local partner that is really homegrown sort of people from there that help us navigate those uncertainties and things that we're not very familiar with"* (BIA-1). The locals can help organizations to navigate uncertainties and help explain things that are not familiar to people living outside of these areas.

The government's role in supporting water treatment systems varies across Sub-Saharan African areas, leading to direct implications on how a company should structure its business model. Where formal government structures are well-developed, they provide a framework water enterprises must follow. In such contexts, formal registration and institutional alignments are conditions a water enterprise is legally permitted to operate. A senior project manager described this structure the following:

> *You have the Minister, the Ministry of Water, then in the county. You have the county government and you have for locations a water service provider. This is like a utility. So it's owned by the county. And then there are community projects, water points which are maybe owned by the community or owned by other stakeholders, other organizations. But then, for example, for the community, you need to register officially as the smallest service providing entity and that is called Water Users Association* (NGO-2).

However, if they are weak, they create both risks and opportunities as this pushes organisations to be flexible to fill the governance gaps that formal structures leave behind. Potential costs of navigating these requirements must be accounted for in the financial planning in the business model. Data highlights the importance of building a solution that is adopted to this situation. By involving locals, a solution can be well established, accepted and taken care of long term. One interviewee shared that "*the street wiser, the street smarter you are as an entrepreneur, I think the better, the higher the chance of success"* (BIA-1). That said, some organizations work very hard with building a strong network with local entrepreneurs that knows the area.

Trust is a factor that emerges from the data as the foundation for project acceptance. Before a local leader gives his approval to any project, trust must be gained, as described here: *"you cannot ask them to trust you. You need to come, you need to involve them [...] eat together, have this kind of, like, human interaction and [...] come with a certain understanding that this will not be a short-term intervention, but you will stay with them in the long run"* (NGO-2). One perspective that was consistently mentioned was how trust is strong within a community, as people tend to trust those they know and are familiar with. When someone enters from outside of that community, they are outsiders, even if they come from the same country. The insider-outsider dynamic may make it more difficult to gain trust from a community. One interviewee explained it more in detail as: "*even us as locals, I come from a different community. So, when I go to these villages, I'm a foreigner"* (LPM-1).

As the data shows physical presence is an important trust-building mechanism and something that is built over time. People in the community need to be involved in the project and organizations need to meet with the people repetitively to build trust. Companies that go to these rural areas need to enter the communities with an understanding that the intervention is not short-term and show the communities that they are there to support them long-term. Another important element that was mentioned throughout interviews was how cultural inclusions directly act as trust signals. Several participants explained the meetings to be open-ended, going into the communities to talk and sit there for hours if needed as: *"that moment of interaction will show that I mean well. I'm not there to take advantage of them. I'm there to do something with them. And it works" (ENT-1*). Many people described this moment as a pre-implementation phase. It becomes an investment in the social infrastructure that determines if a system will be accepted, protected and maintained. The trust mechanism can make or break a project implementation, but also its long-term sustainability. Data shows that the moment there is trust in the community, everything that an implementer says after that will be trusted. This positions trust as an important hurdle to overcome to accelerate a project to succeed in resource constrained areas.

4.1.2 Fostering Local Ownership

Gaining a community's trust does not automatically lead to ownership. Ownership is described as a feeling that a solution belongs to the community and is developed with them and not for them. The data shows that:

> *Many projects set-up, especially here in Kenya, especially in the rural areas where you have NGOs and not-for-profit organizations, they will come and prescribe a solution for a community and go ahead and set it up. So what happens if you do that, you miss a very critical component called ownership, community ownership of the project"* (LPM-1).

The long-term consequence becomes clear, as one local interviewee put it: *"but the moment you leave, they won't maintain whatever you've tried to construct. It's a water facility. And you go there five years later and you find there's nothing because the community actually didn't own"* (GOV-3).

The feeling of ownership can be created in many ways. Data shows that the community needs to be involved in the discussions already in the planning phase. A local interviewee noted: "*so involving the people at the bottom in the conversation. So when you win a bid or a tender, it's high time that you bring some of the people at the bottom to the conversation so that they actually tell you what's on ground"* (GOV-3). It is important that the solution should come from the community, to increase the feeling of ownership. If that is accomplished, ownership is said to create a better understanding and an ambition to care and maintain the product over time.

This need for community-driven solutions becomes important since conditions and needs can vary a lot between communities. In some areas, bore holes are needed, whereas in other communities they have water, but it is contaminated. This finding challenges a fundamental assumption integrated in many organizations' business models, that is that repeatability and standardization can be used to create cost efficiency and enable up-scaling. This was elaborated by an interviewee from a social enterprise accelerator like this: *"if you're the one that sort of is just interested to solve the problem as would be the central actor here, right? You're actually open to say, okay, should it be a water kiosk?*

Should it be piped water? Should it be something different? So I think there is an opportunity for better tailoring the solution to the problem" (BIA-1). In resource-constrained community contexts, these structures are not created for contextual diversity to the extent these areas need. Business models must therefore be designed with flexible and adaptable systems rather than being standardized reproduction. This requirement becomes important for cost structures, implementation planning and adaptation of feasible scaling strategies.

Working together with local community members could potentially path the way for a successful product implementation. One interviewee explicitly shared that: "*there's a lot of what we would classify as indigenous knowledge that can be classified and it becomes a business"* (GOV-1). Another interviewee has a similar experience and shares that:

> *Sometimes the most successful people are the people from the communities and maybe from the former settlements who have never seen a university from the inside, who have never had any idea of how to do a business canvas model or how to fill it up, but they know their community, they know the lack the gap of the source and they know how to, and they have developed solutions for it"* (NGO-2).

While involvement is one very important element to create ownership, inclusion and knowledge sharing is another important part that could pave the way for adoption of a system.

By being given information about the community from locals, project adaptation becomes easier. One example of an unforeseen event is the contaminants level in an existing lake. One interviewee shared: "*maybe it's saltier than you thought, the water is saltier than you thought, and then the whole system, the whole calibration of the system becomes a very complex thing"* (BIA-1). The complexity of an event like that becomes clear when the systems need to be adapted to the new circumstances. The interviewee continued by explaining that: "*probably then the expertise resides with the manufacturer, but the manufacturer doesn't really have clear incentives to sort of make sure that the end user, sorry, the the owner of the system really knows what he's doing because it was not part of the contract"* (BIA-1).

These findings show that long-term sustainability depends on shifting from prescribed interventions towards inclusive ownership. The inherent tension between standardised scalability and contextual diversity leads to the need of business models that favour flexibility over rigid replicability. By recognising indigenous knowledge as an asset and gaining trust as well as community buy-in from the start, organisations can reduce technical risks and avoid the abandoning of projects after handover. Ultimately, success in resource-constrained environments relies on an adaptive framework that links external expertise with local agency, ensuring that the solution is perceived as a community-owned asset rather than something forced onto them.

4.1.3 Accountability Mechanisms and Governance Gaps

The relational and participatory work of the preceding sub-sections must ultimately be translated into formal governance structures to ensure sustainability over time. Assigning responsibility to have someone to hold accountable when the system breaks is a crucial factor for its longevity. The data gives clear implications here: *"a bit of the accountability structure might be an issue, if there is no clear ownership, there is also no clear feeling of responsibility"* (BIA-1).

As already shown in 4.1.2 legal responsibility is often formally clear since mandates often rest at the central government which gets delegated to the local governments. However, data reveals that formal responsibility exists in theory but does not always work in practice. This shows a distinction between formal and informal governance, where actual system functionality depends more on local realities than institutional design. Central governments may assign local governments the responsibility without necessary resources. This creates a gap in the shift as both parties can end up in a situation where they blame each other as neither party feels like they are responsible for the water system. This situation was described by one interviewee the following way:

> *Responsibility and actual capability to offer those services sometimes are not the same, right? So... already with the delegation from the central government to the local government, there's lots of things that get lost in translation, right? So there's an open space that opens for the local government to say, well, okay, you delegated a responsibility, but you didn't assign any resources. So what are we supposed to do, right?* (BIA-1).

This data gives the implication that formal governance arrangements alone are insufficient to ensure successful project implementation and long term functionality. This shows accountability is something that can't be expected when authority and resources are misaligned. The implication for business model design is that resource commitments and defined roles between actors involved are essential. If there is no clear structure, these important elements may lead to project failure.

The mitigation strategy to minimize unexpected problems is to continuously collaborate with multiple stakeholders to gather information on the current status of each community. One interviewee shared how they work by explaining that:

> W*e check the prevalence of waterborne diseases in those communities. And we are going to check the health centers, data, talk to community members. And then after that, if we find, yes, from our own perspective, we feel that there's a need for setting up a project there, and there's a demand for our solution, we will commission a baseline study* (LPM-1).

In each area, it is equally important to establish a clear structure to create responsibility among community members. Unclear responsibility could potentially lead to limited oversight of who is paying to use the system and who is using it without paying. One interviewee explained that, in some cases of piped systems:

> *You usually have little oversight of who's connecting and who's connecting and paying and who's connecting and not paying. So you have sometimes illegal connections, but there's very little oversight again, because there's not enough resources, there's not enough experience, not enough knowledge in the utilities or in the institutions that implement that* (BIA-1).

The findings show that one successful way to assign responsibility at the community level is to look at the accountability mechanisms and to build a system where everyone monitors everyone. One interviewee shared that *"money is kept in a box with several padlocks and no one person has the key to all the padlocks and so they kind of are, they have to be accountable in front of each other to open the box"* (SSS-2). This shows that effective accountability mechanisms can be established from local social structures rather than imported from institutional frameworks. At the international financing level, frameworks such as the Equator Principles, together with tools like Drinking Water Quality Monitoring Plans, help ensure accountability. On top of that additional measures like the following are

taken in bigger projects: *"that is the activity that we have with the local NGOs supporting and also the consultant coming to check in with the selected communities, how it's going to see if there are any trends or any common complaints. We also have grievance mechanisms, which we have designed very carefully"* (SSS-1).

There is a gap between formal governance structures and their practical implementation that shows responsibility is often assigned without the resources or capacity required to fulfill it. This creates an accountability gap and strengthens the misalignment between central and local authorities. In such contexts, unclear ownership further leads to diffusion of responsibility and increases the risk of system abandonment or conflict, particularly when economic value is introduced. At the same time, the findings suggest that effective accountability often emerges from locally rooted actions based on mutual monitoring and social relations and not formal institutional frameworks alone. For a water treatment business model to be sustainable in resource-constrained areas, it must therefore align formal governance structures with local accountability practices to ensure clear ownership, resource allocation and incentives for long-term system management.

4.2 Financial Architecture and Market Feasibility

The governance structures described in Theme 1 are necessary preconditions for a water project to have a foundation. However they are not sufficient to keep it running. As shown in Figure 8 (page 35), once the institutional groundwork is laid, the project faces a second and equally important questions which is how it will be financed, whether it can generate sustainable revenue, whether the market it is entering can actually support it and who will bear the cost of delivering water to those who need it most. These are the questions this theme addresses.

The condition making this theme especially important is a thin margin problem or in other words the struggle to make a profit through the water business. Those who most need water access cannot afford to pay for it and those who can afford it have, in most cases, already solved the problem privately. As one social enterprise accelerator described it: "*maybe our [...] biggest barriers are, I think, is really the money question, not just your end users. Those end users that have the money don't need your product because they already have their own solution*" (BIA-1). This structural gap between need and ability to pay means that financial fragility is something that has to be accounted for from the start.

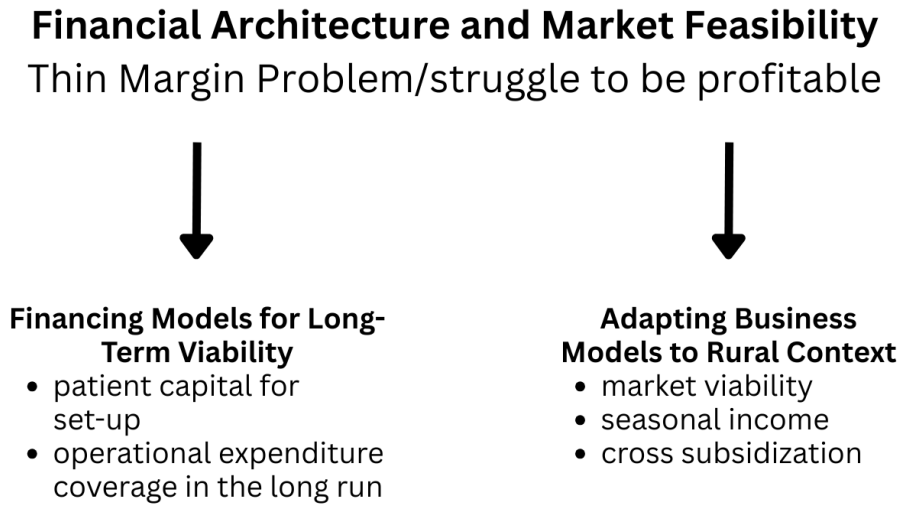


Figure 8: *Financial Architecture and Market Feasibility Theme with Sub-Sections*

4.2.1 Financing Models for Long-Term Viability

Setting up operations in the beginning and entering the rural water market with commercial financing is in most cases incompatible with how that market actually works. Commercial investors require a balance sheet, predictable revenue streams and timelines that the early phase of a water project can't realistically provide. As one local social entrepreneur described from direct experience:

> *In the early stage, it's very difficult, depending on the kind of funding that you get. If you're going to get commercial money, you very, very likely won't get it, because you need balance sheet. Commercial money is looking for balance sheet. They will not finance if you don't have a balance sheet, and if you don't have a positive balance sheet* (ENT-1).

This is not a temporary problem that resolves once the project gains momentum. The first years of a water project in a resource-constrained setting are mainly activities that generate costs before they generate returns. Building community trust, adapting technology to local conditions, establishing governance structures and slowly developing the fee collection habits that any revenue model depends on take time. None of these are attractive to a commercial investor looking for a timely return.

What the data consistently points toward instead for the beginning is the concept of patient capital, meaning financing that accepts long repayment cycles and tolerates the absence of early returns in exchange for the social impact the project eventually delivers. One interviewee from the social enterprise accelerator space described this in the following way:

> *For water and sanitation, that is, you know, oriented towards creating impact and creating universal access. I would say it should be structured along the principle of patient capital. That's probably what you have to, probably true for almost all cases. That's not that with water and*

> *sanitation, as a financier, you can make quick money. Usually you have to have long repayment cycles and you have to be very patient until any of it pays off financially* (BIA-1).

This was confirmed by a local enterprise developer who noted that the sector requires "*financing that is structured in such a way that it's a bit patient. So we really require a lot of patient capital that will be able to handle these businesses until they reach a certain point, then they will unlock maybe debt financing*" (GOV-2). The same logic was explained by a social entrepreneur who had navigated this through grant funding: "*the first three years, it will be difficult to achieve that. So you basically need shareholder money or grant money. And in our case, we got grant funding from EEP just to get us to profitability*" (ENT-1).

Securing the right type of entry financing addresses the first phase. But patient capital and catalytic grants are by definition time-limited. The question that follows is whether the project can build toward financial self-sufficiency before the patient capital runs out. Financial failure is not just occasionally cited as a risk factor in the data, it is named repeatedly as the primary cause of project collapse. As one WASH director put it: "*I think money will always be the number one*" (NGO-1). A local director of innovation and entrepreneurship was equally direct: "*If finances are cut off, it's very, very likely that the project is going to stop. And unless you build a project in such a way that it's viable for the beneficiaries to be able to pay for the project*" (BIA-2).

What the data describes is a structural failure mode that recurs with considerable regularity across the sector. Donor funding arrives, infrastructure is built, the project appears to function and then the funding ends. Rather than transitioning to a self-sustaining model, the project degrades and eventually collapses because no transition was planned for. A water resource authority research officer, speaking from the personal regional perspective, described this pattern in the following way: "*here in Africa, we usually receive donor funding. I think we can start from there, but there is no transition between when the donor funding ends and after the donor funding*" (GOV-3). This absence of transition planning is not simply a project management failure. It reflects a broader culture of donor dependency that the data suggests is reinforced by the repeated experience of replacement rather than continuity. One local WASH project coordinator described an operational mentality that develops under these conditions:

> *What has happened in the past is that we've had a lot of projects done by donors and the people became a bit dependent on donors. And they know, so if you come there, you will set up your project. And after a while, we'll have someone else coming up to set up for their project. So even if it remains, someone else will come tomorrow* (LPM-1).

The unwillingness to pay and take care of systems is made worse where churches provide water for free, where NGOs operate without cost recovery and where politicians promise water as a human right in the lead-up to elections. As the expectation that water should be paid for is difficult to establish in those circumstances and nearly impossible to restore once it has been lost. One senior project manager described:

> *In communities where a church gives water for free, of course, people would not pay for any water anymore. You cannot re-educate them. If policymakers, when the elections are about to come, they go to their villages and say, well, water is a human right, water should everyone has a right to get drinking water, then these people, they count on them, and they say, yeah, okay, it's a*

> *human right, so please give me the water, and I don't need to pay anything for it. And this is also not the case, it's not the truth* (NGO-2).

The implication for any project attempting to establish a revenue model in this environment is that it cannot operate in isolation from the broader sector landscape.

Long-term business planning that accounts for equipment replacement cycles, rather than treating infrastructure as permanent, is a good practice that should be implemented. A local government representative described actively developing this with communities:

> *Will they be able to sustain the plant beyond our presence? [...] they will need to replace the equipment. So they will need to replace the batteries after about 10 years. They will need to replace the solar modules after about 25 years. [...] So then, we help them with a business plan*" (GOV-1).

An international WASH director framed the planning orientation more clearly:

> *I'd be talking to government, I'd be talking to communities, I'd be talking to other organizations in the sector, and I'd be making sure that permeated every bit of our planning and every bit of our thinking, was how are we going to make sure this system functions, not just the day that we cut the ribbon, but five years later* (NGO-1).

The analytical implication is that choosing the right type of entry capital is a design decision that shapes the entire trajectory of the project. A project forced into commercial timelines will, almost inevitably, cut the investments that are hardest to quantify but are important for the long term sustainability, which include trust-building, community engagement and governance development. The profitability of water systems is, as one senior project manager put it, "*really questionable. It is an issue because funders are or banks investors are reluctant to invest*" (NGO-2). Accepting this reality and structuring the entry phase accordingly is therefore not a prerequisite for building anything that lasts. Additionally it is clear that financial self-sufficiency will not emerge naturally from a well-run project in this context. The structural difficulty of covering operational costs from community revenue, combined with a fragmented landscape that actively undermines fee-paying culture, means that self-sustainability must be engineered from the very beginning of the project design.

4.2.2. Adapting Business Models to Rural Context

Some might assume that a viable market exists in all resource constrained areas, but that assumption must be tested before it is made. On top of that the specific context must be understood well before the business model is designed, as the data shows that many projects fail to do this thoroughly before committing to deployment.

Because margins in this context are generally thin, there is almost no room to absorb a market miscalculation. A project that enters on the basis of need rather than demonstrated demand will discover its market problem only after the infrastructure is in place and the capital is spent. Rigorous pre-entry market assessment should be a gate at which the decision of whether to enter at all should be made. As one local government official described it: "*we would want to do then is emphasize more on*

the business - sorry, on the market viability aspect. Are the potential customers able to buy their products? Are they willing to buy that product?" (GOV-1). The same interviewee noted the importance of desk research as a starting point: "*we want to check the economic activity in the region, the general average income levels*" (GOV-1), before any deployment decisions are made. It is important to acknowledge, however, that even a thorough pre-entry assessment can only reduce this risk rather than eliminate it entirely. Markets shift after entry in ways that no assessment can fully predict, which means that ongoing market monitoring is a necessary complement to the initial viability gate instead of something that can be set aside once the initial decision is made.

The buying power question is particularly complicated in this context by a structural paradox that several interviewees identified independently. The segment of the population with sufficient income to pay consistently for water treatment has in most cases already found a solution for their need. Those who genuinely need the product cannot afford it. This is described in the data as one of the main barriers to market viability. As the social enterprise accelerator interviewee put it, those who: "*have the money don't need your product because they already have their own solution [...] and those that need your product cannot afford it. So if you're not super smart and yeah, pull all the tricks possible, then you will probably not succeed because there's just no market for you*" (BIA-1).

Beyond the static question of who can pay today, the data also shows that the market environment itself is dynamic and can in cases shift faster than a project can adapt. Competition in the water technology space exists, with lower-cost entrants disrupting pricing structures that previously appeared viable. A local WASH coordinator in Kenya described this in relation to water ATMs:

> *When they are being introduced, we have only one supplier, and now there are many suppliers of the same ATMs. So you find the cost today is less than half of the cost it was five years ago. But the main supplier has still maintained their costs high. So you find people are moving away and then you find if they are not careful, then they are not going to be able to sustain their business because the demand will be less as time goes on* (LPM-1).

The market-level viability assessment tells you whether sufficient buying power exists in the target population. What it does not tell you is whether that money can actually be collected from individual households, on what schedule and under what cultural conditions. These are separate and in some ways more practical questions, because they determine whether revenue ever actually materialises from a market that appears viable. This is framed not just as a practical challenge but as a question of humility. One WASH director was very direct on this point:

> W*ell, if you come with a European background, you would probably not understand the role of water in a very rural area, village in Uganda or in Kenya. Because the people, they know that water, their water is, it's their life. And so they are very protective of the water they have. And they don't want third parties to come in and intervene or like, take responsibility with the water provision* (NGO-2).

The same concern was raised in relation to external actors who assume they can engineer local market behaviour: "*I'm not convinced that NGOs are very good at economics [...] Nobel laureate economists don't understand the economy, but geniuses who've taken a two-week course then they can manipulate the market and economy of a small town. I'm sorry that's that's that's arrogance*" (NGO-1).

The most fundamental constraint at the household level is seasonal income. A large proportion of rural communities in the contexts described in the data are subsistence farmers whose income arrives once or twice a year, depending on the harvest. A local government official described this reality the following:

> *Much of the population there, they are subsistence farmers. And so their income is seasonal. And it's rain-fed agriculture. There is no irrigation at the moment. So when they plant maize, which is the staple for Zambia, during this period - we have the rain season now - we expect harvesting beginning May, June, somewhere there. When they do that, it dries in the dry season in October. By October somewhere there, we should be - they should be able now to earn by selling to the government, the Food Reserve Agency. And that's the only time they may have tangible income. So we want to be alive to those realities* (GOV-1).

When a farmer is asked to pay a monthly fee for water treatment, the honest answer they could give is: "*look, I earn money twice a year because it's based on seasons. I have my yields of my crops twice a year, so I can pay you then. I can't pay you now*" (BIA-1).

This reality has a direct implication for pay-per-use models, which can appear at first sight like a low-barrier approach to cost recovery. The appeal is logic, rather than asking cash-poor households for a large upfront payment, the model charges only for water consumed. In practice, however, the combination of seasonal income and the operational cost of collecting many small amounts of money from dispersed, cash-poor households makes pay-per-use problematic. Fee collection costs rise when people cannot pay on any given visit and the administrative burden of chasing small payments across a large and geographically dispersed population is itself a significant operational expense. As one interviewee noted, "*fee collection costs is up if people do not answer, people do not pay*" (BIA-1). These two realities, seasonal income and the cost of frequent small collections, speak against pay-per-use as a default model, even where it reduces the barrier to initial access.

There is also evidence that communities themselves manage the equity dimension of fee-based models in ways that externally designed flat-fee structures do not. An international WASH director made the following observation:

> *The communities that I've seen, they know their community, and they know the 80-year-old widow who lives down the lane. They're not charging her the tariff. They're not monsters. The guy running the company? He doesn't care about the 80 year old. So I think the communities that I've seen, they make accommodation for their neighbors. They know who can pay and who can't. And there's other ways to pay or whatever. There's sweat equity, that kind of thing* (NGO-1).

This suggests that community-level collection, rather than individual-level billing, may be more socially effective, even if it introduces its own complexities.

Beyond the inability to pay, there is a separate resistance to paying for water at all as mentioned in the previous section. In many of the contexts represented in the data, water has historically been a free resource, available from rivers and other natural sources. One partial response that emerges from the data is to reframe what the fee is for. Rather than asking communities to pay for water, which they perceive as already theirs, some projects have had more success framing the payment as covering the cost of treatment. As one project manager described:

> *The project needs to make some profits. For making profits, you need to tell the community that it's not about the water. They don't pay for the water itself, because the community already owns the water. But for the treatment, the technical treatment, it's very important that they pay for it properly* (NGO-2).

This reframing does not resolve the underlying economic constraints but it addresses the cultural resistance by keeping the community's sense of ownership of the resource while establishing a legitimate basis for charging for the service.

Another business model element that should be considered is the use of a local entrepreneur for improved maintenance and operations. A paid operator with clear responsibilities and a financial interest in the system's performance is an improvement on the sustainability. When it is built correctly, one interviewee described:

> *There's of course a long-term incentive to provide continuous and professional operational and maintenance services. So that sort of aligns with the interest of your end users. And this is obviously what we always strive to do, right? We have a clear synergy between what the entrepreneur wants and what the customer wants* (BIA-1).

Local ownership of this kind also improves the system's responsiveness in less tangible ways. A local operator can observe and identify risks that a distant organisation would miss and can respond to them before they become failures. As one local WASH coordinator noted:

> *But when you have somebody who is with those localities owning that, then they can be able to observe and they can be able to identify some of the risks they have. And also remedy the same. And it also becomes now the opportunity is maybe it becomes easier for you to scale your business* (LPM-1).

However this introduces fundamental tension. Every intermediary who operates for profit adds a cost layer to a system already constrained by thin margins and cash-poor end customers. The operator who cannot cover their costs from the revenue available will do what any rational economic actor does, they will prioritise the sites and users who generate sufficient return and deprioritise those who do not. As one interviewee stated:

> *If there is not enough money to be made to cover your own expenses, you will also not increase your efforts to improve the service in those areas, and you will focus on those areas that are sort of financially more viable, that is just a matter of survival for the enterprise* (BIA-1).

The result is in the worst case a two-tier system in which those who can pay receive professional service and those who cannot are effectively excluded. As the same interviewee observed:

> *The company prioritizes profit of people, but it just takes its job seriously for those end users that can actually pay for the service, and this excludes... usually the most vulnerable households. So you sort of create a two class or double class society just through the fact that you have a profit driven company*" (BIA-1).

One promising variant to solve this conflict emerges from the data. Namely this is cross-subsidisation from a productive enterprise. Where an entrepreneur operates a business, such as dairy production or a fishing lantern business, that itself requires clean water for its operations, the water system becomes

part of the business infrastructure rather than a standalone cost centre. Community water access can then be provided as a byproduct of the entrepreneur's own operational need, effectively removing the direct dependency on community fee collection. One social entrepreneur described an arrangement that reflects something close to this dynamic in practice:

> W*hen we set up a system, we need water [...] So we will sink a borehole [...] we put some filtration [...] And we make a garden tap available. And so the garden tap is what largely draws the kids around the area, the local community, to come to the center and get fresh water. And if there's a problem with the water, the pumping the the whole setup it's our cost we fix it* (ENT-1).

The outcome, as the same interviewee described it, is that "*we don't charge our communities for water. It's a free resource. They take it*" (ENT-1). A similar logic was applied in a different context: "*the water supply was not profitable enough. So we thought we could cross-subsidize the water business with the business we did with the lanterns, because that was very profitable*" (NGO-2). This cross-subsidisation model is not without its own vulnerabilities. Community water access in this arrangement is dependent on the continued profitability of the entrepreneur's primary business. If the dairy venture fails or the market for the entrepreneur's main product deteriorates, the water system loses its financial anchor and the community might not be able to absorb that loss. This is not a reason to dismiss the model but it is a reason to treat it as one that requires thorough planning and potentially a back up plan that could sustain the water system independently if the primary business fails.

The data makes clear that need and demand are not the same thing and an insufficient market assessment that treats them as equivalent will produce a business model built on an assumption rather than a foundation. Even where genuine demand exists, the combination of seasonal income patterns, high fee collection costs and a deeply rooted cultural resistance to paying for water that has historically been free together constrain which payment models are viable in practice. A purely transactional, pay-per-use model cannot function as the primary revenue mechanism in this context. The income cycle of subsistence farming communities, the operational cost of collecting small amounts frequently from dispersed households are structural features of the environment that the payment model must be designed around. This likely means aggregating payments around harvest income events, operating collection at the community rather than individual level where possible and accepting that reaching the most vulnerable households will most likely require some form of subsidy or cross-subsidisation rather than market pricing alone.

4.3 Technical Sustainability and Operational Resilience

In the previous subsection, it was established that financial sustainability depends on addressing multiple interconnected layers of a margin problem that is based on the resource constrained setting. Even though financial viability is crucial, it is not sufficient on its own. A system can remain financially sound and still fail if the technology breaks down and no one can repair it, if trained operators leave for better opportunities, if spare parts cannot be sourced locally or if seasonal flooding cuts off site access. These subsections focus on the technical dimension of sustainability, the operational resilience of the system once it is operational. The key insight across the three sections shown in Figure 9 on page 42 is that technical sustainability is more than a technology problem, it is also a people and incentives problem. The hardware itself is often the most manageable part of the challenge. What is more difficult is ensuring that the right people are present, trained, motivated,

equipped and reachable over the long term. Each section examines one layer of this challenge and each layer must function effectively for the system to remain operational after the implementing organisation leaves the site.

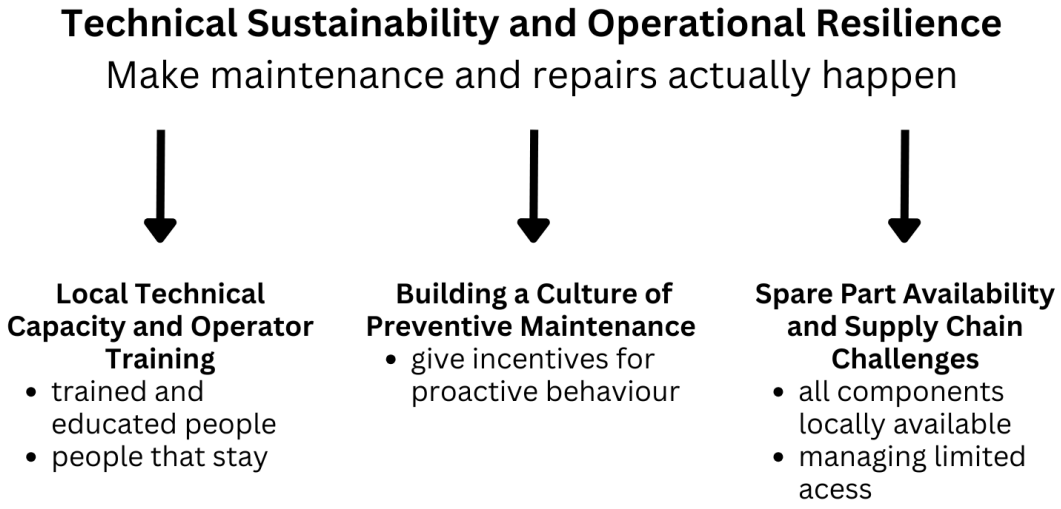


Figure 9: *Technical Sustainability and Operational Resilience Theme with Sub-Sections*

4.3.1 Local Technical Capacity and Operator Training

Where functional literacy is limited, community members struggle to participate in governance, follow written protocols, manage financial records or hold service providers accountable. One senior sustainability analyst described encountering this barrier the following: "*examples of challenges that we've had is, for example, that some people who don't read and write, they feel that they would not be able to engage, so it was difficult to find people*" (SSS-1). The issue is not a lack of commitment or interest but rather that the operational model is in cases designed with literacy in mind and assumes a level of education that is not necessarily present. This creates a gap between what the model requires and what the community can provide.

Beyond literacy, the data highlights another critical factor, institutional open-mindedness or the community's ability to engage with new practices, technologies and forms of organisation without dismissing them outright. A local enterprise developer summarised this connection the following:

> *Education to a certain extent gives you the ability to be open-minded to a certain extent and not to be lack of a better term, dismiss genuinely important things that are there. If you are educated, you are open-minded, you have a skill, you are better placed to actually try and start a business, do that skill and progress further* (GOV-2).

This matters not only for adopting new technologies but also for the behavioural change required for sustainable water use.

The data also shows that education within the leadership layer can be disproportionately impactful. When traditional leaders or local government officials understand a project's requirements and can communicate its value from a position of knowledge and trust, implementation becomes significantly simpler. One interviewee described a case where the traditional leader in their project area had previously been a school leader: "*he was very alive to what was required of the project. And especially that the relationship between the traditional leaders in Zambian government is very healthy*" (GOV-1).

After looking at the community's collective capacity to engage with and govern the water system, it has to be made sure designated individuals are actually empowered to fix broken water systems. The choice of technology directly determines what local technical training can achieve. A system relying on proprietary components, complex electronics or imported-only consumables creates a maintenance dependency that no amount of training can overcome, because the parts or knowledge required are simply unavailable locally. On the other hand, a system designed with local serviceability in mind can significantly expand the pool of people who can maintain it. The chief marketing officer of the water tech startup described their design approach this way: "*we've also taken great care to make sure that everything is serviceable at the lowest possible level, ideally by the user themselves. So any part can be interchanged with simple hand tools by basically anybody who is able to use hand tools*" (COMP-2). The company noted another design principle: "*we tried to... use as much standard equipments and standard solutions as possible so that if for some reason we wouldn't be able to deliver, they can find most of this stuff locally.*" (COMP-2). Technology and training must therefore be treated as interdependent decisions made together.

Given a well-designed technology, the training structure supported by the data is split up. Basic operation, routine maintenance and first-level troubleshooting are covered by face-level training. Delivered by the implementing organisation or its local partners. Deeper technical training, requiring more specialised knowledge or equipment, is provided through external partners or regional training centres. A local enterprise developer described this division of responsibility: "*we do face level training that we can do ourselves, we do train, but technical more deep training, we do work with our partners and our stakeholders to do that*" (GOV-2). A concrete example of this approach was shared by a local government official who arranged for four young men from a rural village to spend three weeks at a regional training centre:

> *We identified four young men from the village whom we brought to Lusaka and took them down. [...] They stayed there for about three weeks or so, where in theory, they went into the labs. And then they've gone back to the village what follows is they will do hands-on practical experience as being set up so those are being trained to operate and maintain the plant okay so that will ensure that uh the plant is available 24/7* (GOV-1).

This combination of theoretical instruction and practical application, followed by local hands-on experience appears to be an effective solution.

Ideally the trained people handle the operations later as their main job, as it is a lot to ask of the local community members to handle this on the side. One interviewee from the social enterprise accelerator space put it this way:

> *These are people who are not professionals they're just normal citizens like you and me and then they have some sort of decision-making power in that community but they sort of have a job at the same time and have 10 other things that they have to attend to and then they should become they should be responsible for professionally maintained and operated water and sanitation infrastructures, which is just a very tough thing to do* (BIA-1).

This is not an argument against local technical resilience but it is a reason to be realistic about the support structures required.

Identifying suitable trainees is itself a challenge. Selecting individuals who have both the baseline capability and the genuine willingness to stay and apply their training requires attention to community dynamics. Local leadership involvement in the selection process is described in the data as both practical and culturally important. One interviewee noted: "*the emphasis is that you use those that are from the area, not someone from the nearby town*" (GOV-1), a principle that links local hiring directly to retention. The cost of ignoring this is significant. As one local director of innovation explained: "*if you are sending somebody from the hub, from the key, from the center to the node, every single time that you have a breakdown, then that can be expensive and that can also take time*" (BIA-2).

The selection of the right trainees for the investment in local technical training just described is especially important, as the technical abilities increase the employability of those trainees, making it more likely they will leave. This is not a fault of character or commitment on the part of those who leave. It is a rational economic response to a labour market where trained individuals command value that a water project, constrained by thin margins, often cannot match. One documented version of this dynamic in the data concerns competition from NGOs. Larger international NGOs typically offer salaries and benefits that small social enterprises and community-level operators simply cannot compete with. One interviewee from the social enterprise acceleration space described the pattern the following:

> *We see a lot of brain drain as well with social entrepreneurs, social enterprises compete with NGOs for the best people. And usually NGOs, at least the bigger ones, are able to pay very good salaries [...] So there's a sort of a tendency for brain drain from social enterprises, especially the younger ones, the smaller ones that are not able to pay very high salaries, and they usually struggle to compete against bigger NGOs, right? So it's also tough to, you know, perform the expected way if your staff is just difficult to train, or once they're trained, they just go to the next bigger NGO who pays better* (BIA-1).

It is worth noting the connection to the literature in 4.2, the financial constraints that make it difficult for social enterprises to cover operational costs also make it difficult to offer competitive salaries. The operational consequence is a loss of technical capability potentially right after the investment in this capability was made.

Even where NGO competition is limited, the general movement of young, skilled people toward urban centres is a consistent feature of the landscape that has to be kept in mind. One social entrepreneur described this directly:

> *So it's a lack of skills in that area, in those locations. We can't find skilled people that we can employ that live there. So we have to rely on young people coming from Lusaka. The thing about young people also is that they don't want to live in the rural areas* (ENT-1).

Another interviewee with training and commercialisation responsibilities emphasised the same point:

> *You never know what happens to the people that you train. Especially if you, there's always the risk of as soon as you train people, they become employable and they become employed by someone else that lives far away or they suddenly see, okay, now I have a chance to actually find a job in the city and they just move out* (BIA-1).

Prioritizing trainees who are deeply embedded in the community and attached to it and therefore less likely to leave is perhaps the most feasible mitigation strategy. One local official framed this as follows: "*for those that are in the rural areas specifically[...] they are too attached to the area however the risk is not very high*" (GOV-1). A complementary approach is to distribute training across multiple individuals rather than concentrating it in one. Where a single person holds critical operational knowledge and that person leaves, the system is effectively deskilled overnight. Where knowledge is distributed, the departure of any individual does not mean a catastrophic loss. This connects to the broader point made by a water resource authority research officer:

> *If you don't have the capacity, not only the financial capacity, but the human resource capacity to actually maintain it, because it's one thing to have very good infrastructure and the money to power it, but it's another thing to have people actually who know how to use it and that delivering the outputs of which that you actually set the machinery for. So I think human resource capacity is as equally as important as financial capacity. So if you don't have the human resource capacity and you can't outsource, then it's a failed project at the end of the day* (GOV-3).

It is also worth noting that the brain drain problem is not limited to the operational level. The same vulnerability applies at the enterprise level. Where a project depends heavily on the specific knowledge, relationships or vision of a founder or key technical lead, the departure of that individual can destabilise the entire enterprise. As one local enterprise developer observed: "*The second factor is usually founder dynamic. Whatever happens between the founder and specific founders as well. Sometimes it might not necessarily be founder dynamic, but some form of like founder continuity strategy of some sort. So if one of the main co-founder leaves the company, what follows next*" (GOV-2). Succession planning and knowledge documentation are therefore needed.

This data suggests that the educational baseline of a target community acts as a design constraint, operational models must compensate for with simpler protocols, more intensive support. A business model that assumes consistent community engagement capacity across contexts is likely to encounter problems that could have been anticipated and designed for. The evidence further shows that local technical resilience is based on a deliberate alignment between technology complexity and local capacity. A sophisticated system deployed in a community without the skills to maintain it will degrade just as surely as a system with no maintenance budget, even though maybe more slowly and less visibly. Technology as well as trainee selection and training programme design must be treated as a single decision. Trained capacity that is not retained is practically a contribution to the wider labour market rather than an investment in the system. Therefore distributing knowledge, selecting people with strong community ties and if financially possible, offering genuine incentives for staying are among the most important elements.

4.3.2 Building a Culture of Preventive Maintenance

At a certain point, the question of technical sustainability shifts from people's capabilities to the habits and incentives that determine whether those capabilities are actually applied. The previous section addressed what training is necessary and how to have the capabilities located where they are needed. This section focuses on what they do and whether the institutional environment around them makes proactive maintenance a realistic daily practice for better sustainability.

The cultural starting point matters more than it might be recognised in technical deployment discussions. In the context represented in the data of this case, the concept of preventive maintenance, which is intervening in a functioning system to prevent future failure, is to some genuinely unfamiliar. A local WASH coordinator made this point:

> *I have to mention, that is one of the things that we have struggled with, preventive maintenance. You know, I have checked and locally, and we have many languages in Kenya, we don't have a local word for maintenance. So which means we are not used to maintaining it. So if it is working, why open it? So for us, we wait until the day it is completely broken to start with the repair* (LPM-1).

This reflects a relationship with infrastructure that differs significantly from the assumptions made by providers that are not familiar with the culture. It cannot be addressed through a training session or a written checklist. It requires a sustaining cultural shift.

Even where the concept of maintenance is understood, the absence of financial incentives means it will not happen reliably. Maintenance is by its nature invisible when it works. No one notices the leak that was caught early or the filter cleaned before it is blocked. The result is an underinvestment in preventive action, which accumulates until a breakdown makes the cost visible and immediate. The design response that emerges most clearly from the data is to make responsiveness financially rewarding. One senior project manager described a specific and apparently effective incentive structure: "*the incentive here is for the local team that whenever the leakage or break of a system is reported that they get a full amount of money they've agreed upon. If this is fixed within one day and then it gets lower the longer it takes to fix the system*" (NGO-2). This structure directly rewards proactive behaviour.

The question of who is responsible for maintenance is addressed consistently in the data through a preference for dedicated, paid employment rather than part-time or volunteer arrangements. One local WASH coordinator said: "*for our project, it has been very clear that we need to have a full-time employee for that facility*" (LPM-1). An export finance director also stated the importance of assigned responsibility: "*it's very important that the operations and maintenance is someone is responsible for that and reliable revenues covering all the costs*" (SSS-2). Without a named individual accountable for the system's condition and without external monitoring to verify that accountability, deterioration can go undetected until it ends up as a complete breakdown.

Community ownership can serve as a different and complementary form of protection. Where the community has a stake in the continued functioning of the water system, infrastructure maintenance and physical protection become a collective interest rather than an assigned responsibility. A water resource authority research officer described a case where community members were engaged to

collect hydrological data and did so with notable commitment once they understood the direct benefit to them: "*they are collecting data for us. And there are incentives, and they are so committed, you know, like they take it like they are personal properties*" (GOV-3). The key enabling factor, as the same interviewee explained, was that the community understood the data they collected would directly reduce water scarcity in their own areas:

> *Even when now we came with the concept of them collecting the data and so on, they were very receptive because they understand that this data we're collecting is not just for scientific purposes, but it's actually going to help us have, you know, shared equity among water, you know, allocation within their areas* (GOV-3).

The reverse is also true. Where community ownership is absent, maintenance culture suffers. An international WASH director noted that communities can be actively hostile to private company involvement in water management: "*it's not unheard of for people to start rioting when they hear about private companies. And the bigger the company, The more criticism you're going to get*" (NGO-1). Maintenance culture here is not a technical matter at all, it is the practical side of the governance and trust relationships that Theme 1 describes as foundational. Communities that were genuinely involved in the design and governance of their water system are more likely to feel that it is theirs to protect.

This section suggests that maintenance culture is built through education, ownership and incentives working together and that each is limited without the other. Financial incentives for responsiveness are necessary but work best when they reinforce an existing sense of responsibility. Community ownership is a powerful driver of protective behaviour but on its own does not produce the systematic attention to preventive maintenance that a water treatment system requires. Where both are designed for, the chances of long term sustainability are significantly increased.

4.3.3 Spare Part Availability and Supply Chain Challenges

Even a well-incentivised, community-embedded operator who takes genuine pride in the system's performance cannot maintain equipment that cannot be repaired. The availability of spare parts, at the right location, at the right time and at a price the community can afford is a precondition for long term functionality that might sometimes be dismissed as a logistical side note but is in reality a fundamental design decision that has to be accounted for from the very beginning. As one WASH director put it: "*it really does come down to that sustainability snapshot that I shared with you [...] Does the community have money? Do they have spare parts? And do they have technical support?*" (NGO-1). These three questions are interdependent, but spare parts availability is the one most directly shaped by the technology selection decisions made at the project's beginning. It is also worth noting the connection to the financial aspects here, communities must have sufficient fee revenue to purchase replacement parts, which means the spare parts question and the payment model question are not independent of each other.

The most critical vulnerability in this domain is the use of proprietary or import only components. A technology that relies on filtration parts or electronic elements that can only be sourced from the original manufacturer or imported from abroad places the operational sustainability of the system at a

dependency of supply chains outside of the community's or even the country's control. A WASH director highlighted the risk:

> *At the end of the day, if you've got a technology with proprietary elements, either the fitting, parts in the thing, or the necessary reagents or chemicals in it, all of those things, anything that's proprietary and coming from outside the country, the chances of that failing are very, very, very high* (NGO-1).

The failure mode is not necessarily immediate. The system may run perfectly for several years before a critical component fails and cannot be replaced. The same interviewee raised the risk of policy-level disruptions to import supply chains:

> *I think that all technology has to be from there or all products but you're really running the risk if there's not enough purchasing power, and they've just stopped bringing the reagents in, there's some new government policy, somebody has the bright idea of raising taxes on outside reagents or whatever, that all of a sudden the price doubles and oops, the system doesn't run anymore* (NGO-1).

This is not hypothetical in contexts where government policies on imports and tariffs can shift with little warning.

The design response supported by the data is to treat local sourcing as a primary criteria. The wWater Tech application for an EU project acknowledged this risk directly: "*Limited local supply chains for technical components (UV, ozone, solar) could raise costs or delay delivery*" (Water Tech Startup EU Project Application). The chief marketing officer from the same company described the design philosophy that followed: "*we tried to... use as much standard equipments and standard solutions as possible so that if for some reason we wouldn't be able to deliver, they can find most of this stuff locally*" (COMP-2). The startup's pitch deck summarises this commitment as being "*built with standard, globally available components for long service life*" (Water Tech Startup Pitch Deck).

Another factor that needs to be considered in this context is that road infrastructure in many rural contexts described in the data is highly variable and sometimes simply impassable. One social entrepreneur offered a direct perspective of the operational consequences: "*now that it's been raining the last two months, our volumes have completely reduced, maybe by almost half. Because there's so much water. The roads are impassable. And farmers are not able to work in some cases, because it's just flooding*" (ENT-1). The same interviewee described having to close one service centre temporarily: "*the road infrastructure, is really bad. And after it's rained so much, the road is impassable. So we may have to shut that one down until they're fixed, then we can come back*" (ENT-1). These are not extraordinary events, this can happen regularly. Any operational model that requires regular external access is at a higher risk for that reason. Pre-stocking critical components within the system itself is one response. The water technology company described:

> *We also have a prepared spare parts kit with those components that have a shorter lifespan than 10 years, so that you already have a spare part in the container. So that if something breaks, you can easily just change it again and out, get it going again, and then order, not having to wait for the part to arrive* (COMP-2).

This approach reduces downtime but doesn't eliminate the whole underlying supply chain dependency.

The presented evidence here suggests that spare part supply chain design is a sustainability decision made at the point of technology selection, not a procurement decision made after deployment. A technology that cannot be maintained with locally available components is not appropriate for that context, regardless of its technical performance when new. The business model must therefore include an explicit mapping of what components are likely to fail, at what intervals, where replacements can be sourced and at what cost. On top of that, operational models for remote settings must be designed from the outset to function with minimal required external access, because that access cannot be guaranteed.

Across the three sections of this subsection, a picture emerges. Community-level education provides the governance foundation. Operator-level technical training builds the capability to run the system independently. Retention strategies keep that capability embedded locally. Institutionalized maintenance ensures that the capability is exercised proactively. Local spare parts availability means that when something fails, it can be fixed without waiting for external intervention. And the physical inaccessibility of the site in some circumstances reinforces all of this. The practical implication for business model design is that each of these layers must be considered and that the weakest one typically determines whether the system is still running ten years after its implementation. What the technical part of the thesis shares with earlier parts of the report is the same fundamental logic, neither financial nor technical sustainability is the default outcome of a well-intentioned project. Both require deliberate planning.

5. Discussion

This study focused on understanding what elements need to be considered for a business model of an off-grid water treatment system to be sustainable in a resource-constrained area. The answer to that is:

Multiple elements are required for a SBM of an off-grid water treatment system to be sustainable in resource-constrained areas. Sustainability does not happen by accident, it must be intentionally designed into the business model from the project's earliest stages. For these systems to function effectively over time, different interdependent elements must be present.

Trust is fundamental, as communities without trust in implementers refuse to pay for the system, which leads to a lack of maintenance. Financial stability is equally crucial, since payment enables maintenance, while financial instability results in long downtimes and eventual abandonment. In addition to that, maintenance depends on both good governance and reliable financing. The discussion will examine the relationship between these findings and existing literature, as well as their implications for practice.

5.1 Community, governance and institutional design

The findings both confirm and deepen the review of the SDA. Existing literature has discussed that long-term functionality for water systems depends on professional management and lifecycle support (Lockwood & Smits, 2011; Moriarty et al., 2013). On the other hand, literature is less clear about how these types of support should be organized in reality. The findings of this thesis contribute to that gap by providing empirically grounded examples of the organizational dynamic that can support sustainability long-term. These elements consist of trust-building, together with accountability structures that are shaped through local social norms. One more element that is important is the support from traditional leadership before implementation. The thesis does not resolve the SDA debate but it offers insight into how these organizational structures function and may operate in practice.

Trust and social capital have to be seen as underlying conditions for sustainability instead of being seen as relatively soft factors. Literature by Harvey and Reed (2007) and Hutchings et al. (2015) discuss that trust influences whether communities continue to support and maintain water systems over time. The findings justify this argument with information about how trust is built through physical presence, interaction and cultural inclusion in the community before the system is implemented. Once that is established, trust functions as a bridge that strengthens participation, WTP and long-term engagement with the system.

The importance of traditional leadership connects the findings to legitimacy theory discussed in the literature review. Fisher et al. (2017) argue that organizations gain acceptance through alignment with local actors, while Suchman (1995) describes legitimacy as both moral and cognitive acceptance. In the findings, approval from traditional leaders functions as a key signal of institutional credibility and social acceptance. Without this support, even systems with good technology are at risk of being resisted or abandoned.

Information about liability of people that are considered to be outsiders is added through this work. The data shows that even actors from the same country may still be perceived as outsiders when operating in unfamiliar communities. This highlights the importance of being consistent with local engagement which has direct implications for partner selection and business model design. Local presence should therefore be treated as an ongoing operational requirement rather than a temporary project phase.

There is information in the findings that confirm critiques of the gap between formal governance responsibilities and actual resource allocation. Van Den Broek and Brown (2015) describe how systems often fail when responsibilities are assigned without proper financing or resources. Similar patterns are identified by Kativhu et al. (2018) in their study of community-based management in Zimbabwe, where local actors were formally responsible for water systems but lacked the resources and capacity to fulfill these roles. This thesis extends this critique by showing that the problem remains in business models of different kinds. Effective governance design therefore requires alignment between responsibilities, resources and institutional capacity rather than only assigning roles on paper.

Ownership is shown to be created through participation in the process rather than through formal structures alone. Findings showed that community members were more likely to invest in, maintain and protect systems when they had been involved in planning and decision-making processes early in the project. At the same time, systems that are introduced without meaningful local participation were more likely to be neglected or abandoned. This supports the arguments of Gedamu et al. (2025) and Hutchings et al. (2015), while also identifying the specific organizational practices that strengthen ownership. That includes early-stage participation and the integration of indigenous knowledge throughout the planning process.

5.2 Financial Architecture and Cross-Subsidization

The findings related to financial architecture represent the central analytical contribution of this thesis. The data confirms what can be described as a paradox based on a thin-margin problem. The communities that are most in need of safe water often have the lowest ability to pay for it. Those with enough financial capacity already have secured alternative water solutions. This creates a gap that makes long-term sustainability difficult for any model that relies heavily on fee collection or tariffs. Existing literature identifies the financial tension, but provides limited explanation of how it can be resolved in practice (Bocken et al., 2014; Onsongo et al., 2023). Looking at the data it suggests sustainability requires organizational arrangements that go beyond individual user payments as the primary revenue source. One of many interesting findings is therefore the role of cross-subsidization. From the information collected through interviews, water systems became more sustainable when they were connected to an organization that is dependent on clean water for their own operations. In these situations, community access to water became a secondary benefit of a commercially viable activity instead of depending entirely on local tariff collection. This makes the financial aspect less heavy for individual households to get access to water.

Two cases illustrate the financial aspect in relation to a water system. The first is ENT-1’s model, where a dairy business and a borehole system were financially linked. The second is NGO-2’s lantern

business, where revenue from one activity supported community water provision. These examples connect well to the literature on frugal innovation. FI supports multi-actor collaborations that distribute costs, reduce financial pressure and support long-term adoption (Onsongo et al., 2023; Rosca et al., 2017). This thesis provides a concrete example of how such principles can operate in practice. From a business model perspective, these structures also meet the criteria for Business Model Innovation outlined in the literature review. Using the Magic Triangle framework (Gassmann & Frankenberger, 2025), at least two parts of the business model are changed. Those two are who the main customer is and how the company makes money. Revenue is no longer only connected to household water payments, but to a broader commercial ecosystem that indirectly sustains community access.

The findings can also be interpreted through the NICE framework discussed by Zott and Amit (2010). The cross-subsidization model creates a link between the entrepreneur's commercial incentives and the community's need for water access. In this way, sustainability is supported through incentives that are built and aligned into the business model itself. It does not only rely on good intentions, being dependent on donors or other external enforcements. At the same time, it's also shown that cross-subsidization models also contain vulnerabilities. Community access to water remains dependent on the continued viability of the entrepreneur's primary business activity. If the business fails, the water system could potentially lose its financial base. For this reason, the findings suggest that backup financing should be treated as one main design requirement instead of an optional safeguard. This issue should therefore be highlighted as a practical design recommendation. However, since this argument is based on only two empirical cases, it should also be considered together with the limitations discussed in section 5.6.

In addition to this it is shown that patient capital is a structural requirement for water projects in resource constrained areas. As mentioned in several interviews, respondents argued that conventional commercial financing is poorly suited to the early phases of rural water initiatives. This aligns with the OECD three T's framework discussed by Lockwood and Smits (2011). They argue that tariffs, taxes and transfers are all necessary because no single funding source is sufficient on its own. The findings extend this argument by showing that financing structures shape the development trajectory of a project. In any attempt to meet short commercial timelines many times result in reduced trust and community engagement and governance development. Those cannot be forgotten as they are important conditions required for long-term sustainability.

Seasonal income patterns were also discussed in the result as a challenge to conventional tariff systems. Monthly fee collection models were repeatedly described as poorly aligned with local economic realities as many household incomes are dependent on agricultural cycles. This finding is supported by previous research where both Chowns (2015) and Kativhu et al. (2018) identify seasonal agricultural income as a key reason why monthly payment systems often fail in practice. More information was gathered from Gedamu et al. (2025), who address irregular payment capacity to both poverty and unstable incomes. This leads to the conclusion that harvest-based payment systems or community-level collections may be more viable than individual monthly billing systems. The data also shows that how water fees are collected matters culturally. In the result section, it was described how communities were more accepting of payments when fees were presented as payment for water treatment services rather than payment for water itself. However, while reframing could potentially reduce cultural resistance, it does not solve the broader structural issue of limited and irregular

household income. Both cultural legitimacy and financial alignment are therefore necessary conditions for sustainability.

Finally, the findings touch on concerns regarding donor dependency. Several respondents described how prolonged reliance on donor funding weakened both government accountability and community expectations around payment and maintenance responsibilities. This pattern is also well discussed in the literature. Gedamu et al. (2025) and Kativhu et al. (2018) both argue that being dependent on donor funding can undermine long-term institutional responsibility and reduce incentives for locally sustainable financing systems. Therefore planning for a gradual transition from grant-funded entry phases toward self-sustaining revenue generation should be considered a central business model design requirement and not a secondary implementation issue.

5.3 Technical Sustainability as a People and Incentives Problem

Technical sustainability should not primarily be understood as a technology problem, but to be connected to people, incentives and organizational design. The WEIHN framework discussed by Huang et al. (2023) provides a useful conceptual foundation for interpreting these dynamics. The data collected add empirical depth to the framework's human dimension. Rather than treating human factors as a single category, the data indicates that they should be understood as three distinct but connected challenges. The challenges are connected to having the right capacity, retaining the system and establishing a maintenance culture. Capacity concerns are referred to if people can properly operate and repair the systems. Retention focuses on whether trained individuals remain in the community, while maintenance culture is connected to how preventive maintenance is valued and consistently carried out over time. Each of these dimensions requires different forms of intervention. While this thesis does not formally test the WEIHN framework, it contributes empirical insight into the complexity of human connections.

One important finding is that brain drain should be understood as a predictable outcome rather than a failure. Training increases technical competence and employability, which can create strong incentives for skilled individuals to leave rural areas for better opportunities in another area. The findings in this thesis suggest several possible responses to this challenge. One includes distribution of technical knowledge between multiple trainees and selecting operators with strong local ties. Where it is financially possible, companies could provide incentives to support long-term retention. Importantly, these responses should not be viewed as operational decisions but as business model design choices with financial implications.

Another important finding is that preventive maintenance is an important cultural and organizational challenge. One respondent shared that many Kenyan languages do not have a direct word for maintenance. This could potentially indicate that many water projects are built around assumptions about culture towards infrastructure that do not fully match local experiences or everyday practices. The issue connects to Carter and Ross (2016), who discuss that looking at rural water systems as simply functional or non-functional overlooks the more complex reality on the ground. According to their perspective, sustainability should instead be assessed through factors such as repair history and downtime duration. It is shown that technical sustainability depends on more than only on training operators. Creating incentive structures that encourage preventive action before breakdown occurs is

another important point that needs to be considered. In this sense, maintenance culture must be part of the business model design rather than assuming it to emerge naturally. One example is NGO-2's tiered payment structure, which encouraged proactive maintenance instead of only responding when something broke. This shows how organizations can use incentives to support behaviors that help strengthen long-term system functionality.

The availability of spare parts is another topic that has to be considered. The data suggests that systems depending on specialized technology or imported components are especially exposed to supply chain disruptions, changing regulations and weak transport infrastructure. This leads to the conclusion that spare part access should be treated as a central criteria in technology selection rather than being a secondary procurement issue. This interpretation is supported by Lockwood and Smits (2011), who discuss that sustainable rural water management depends on technology standardization to create more stable markets for spare parts. They also highlight that NGO-managed spare part stocks are many times temporary because they often disappear once projects end. The findings of this thesis support this argument by showing that dependence on certain technologies can create long-term vulnerabilities, even when the initial implementation is successful.

The OWTS design can be understood as a practical response to these combined empirical and theoretical challenges. By focusing on standardized components and available spare parts to enable locally repaired systems, the business model could reduce reliance on external suppliers and improve local repair capacity. In this way, technical sustainability depends not only on the technology itself, but also on how well the system fits local organizational, financial and maintenance realities.

5.4 Societal and Ethical Implications

Multiple societal and ethical implications of the findings can be identified. One key concern is the risk of two-tier access under profit-driven operator models. The data shows that vulnerable households may be deprioritized when they cannot cover operational costs. This is not an isolated observation. Chumo et al. (2025) and Sarkar (2020) document similar dynamics in informal water markets, where profit motives can lead to exclusion of less financially viable communities. Business models that do not address this issue may contribute to exclusion. Cross-subsidization can serve as one mechanism to address this. Accountability structures that are aligned to serve also the most vulnerable offer another.

The donor dependency presents another crucial ethical consideration. The data suggests that repeated build-and-leave interventions can erode fee-paying cultures that can finance water infrastructure. This can make subsequent projects harder to sustain as they have to overcome this additional challenge. Gedamu et al. (2025) and Kativhu et al. (2018) describe similar patterns. Organizations designing business models for this context need to take steps to avoid repeating this cycle, for example through choosing approaches like cross-subsidization that can reduce the fees in general.

The human rights paradox also appears in the findings. Water can be seen as both an economic good requiring cost recovery and a basic human right, creating tension around mandatory payments. The fee-reframing strategy, where communities pay for treatment rather than water, seems to offer a practical way to navigate this tension, as indicated by the data. However, it may not fully resolve the underlying ethical dilemma on how water should be treated in such resource constrained areas.

Gender dynamics surfaced in the coding as another ethical implication. However, these were not developed into the findings, as they did not relate directly to the elements that improve the sustainability of a water system. More on this on the limitation is 5.6.

5.5 Evaluation of Methodology

The single-case design can be considered appropriate for theory generation in an under-studied organizational context, as suggested by Ketokivi and Choi (2014). The idiographic focus was an intentional choice, with statistical generalizability being a limitation resulting from this. Transferability is supported through rich description for comparable resource-constrained contexts, as recommended by Bell et al. (2019).

The pre-commercial status of the OWTS means that findings are grounded in practitioner experience with similar deployments and their expert judgment, rather than post-deployment evaluation. Claims about the OWTS specifically, such as its design philosophy and spare part approach, are based on company documents and interviews, rather than field observation. This limits the strength of the confirmation of claims about how those design choices will perform in practice.

The Gioia methodology and the interpretive leap from first-order to second-order themes involve researcher judgment, as outlined by Gioia et al. (2013). Constant comparison was used to help limit selective quoting and both researchers conducted coding to help improve confirmability. This process is described in section 3.6. However, researcher bias can be a limitation that persists despite deliberate efforts of eliminating it.

5.6 Limitations

The following limitations and implications should be noted alongside the ones already noted in the methods chapter. One consideration is the collaboration with Water Tech regarding confirmability and to the limitations this single case brings with it. The company's interest was in an unbiased analysis. The data collection and analysis, besides the interviews with members of the company, were independent from company involvement. Additionally as described in section 3.8, all interviewees were informed about the collaboration.

Another limitation concerns the cross-subsidization finding. This model is supported by two concrete cases in the data, which may represent a relatively thin empirical base for a strong theoretical claim. Further research with more cases and post-deployment data could help establish boundary conditions, such as at what scale, in what primary business types and in what governance contexts the model holds.

During the sampling process the aim was to also include end users who live in the communities to get their view, as they are arguably the most important stakeholders. However, all tries to reach out to end users and water user committees were without success. While the collected data does not not suggest that any findings could be challenged if they were included, it is always a chance that their perspectives would have altered the findings of this thesis. So this limitation should be kept in mind.

A further limitation relates to gender dynamics. These surfaced in the coding but were not developed as a finding in Chapter 4. This has been a decision made based on the scope of the thesis. The data suggests gender can influence both governance participation based on the culture in those areas and resource allocation within the families when they receive payments. Those can be seen as factors a business model can address to improve the equality and economic situation of families in resource constrained areas but they have little direct impact on the long term functionality of a water system.

6. Conclusion

Rural water systems often don't fail from a lack of technology or initial funding, but from how responsibilities, incentives and financial flows are organized around the hardware. This thesis asked what elements of a business model support the long-term sustainability of an off-grid water treatment system in a resource-constrained area. The findings suggest that sustainability has to be intentionally designed across three interdependent domains, none of which is sufficient on its own.

6.1 Thesis Contribution

Long-term sustainability requires the simultaneous presence of community governance, financial architecture and technical resilience. These three domains are interdependent. Trust makes people willing to pay for water. These payments fund regular maintenance. Effective maintenance then depends on both clear local oversight and financial stability. A failure in any single domain can lead to the failure of the whole system.

In governance, a physical presence and cultural respect build trust before the system is even installed. Gaining support from traditional leaders is a necessary start point. Local people should help design the system so they feel they own it and accountability only works if formal responsibilities match the actual resources available at the local level. The financial structure requires patient capital during the early phases to engage in the necessary community governance. Using a profitable business to cross-subsidize water costs looks to be a good way to recover expenses, when the most vulnerable groups are not able to pay for it themselves. Payment schedules should match agricultural harvest cycles rather than using rigid monthly bills and the transition from grants to self-sustaining revenue must be part of the plan from the beginning. Technical resilience relies on using standard parts that can be fixed locally by local technicians. Training multiple operators helps the system survive if one person leaves, which has to be accounted for because trained technicians and operators become employable by others. Incentives should reward preventive care before a breakdown happens, as this behavior is not standard behaviour in this context and therefore must be actively promoted. Keeping spare parts on-site is a way to enable repairs even if the area is temporarily cut off from the infrastructure. Using only some of these elements will likely result in the same failure patterns the sector has seen for decades.

This research bridges the gap between water studies and business model theory. Water research identifies why systems fail but often lacks the design tools to fix them. Business model theory provides those tools but usually assumes markets work perfectly. This thesis applies these tools to settings where contracts and profit incentives are less reliable.

The cross-subsidization model is the key result and contribution of this thesis. It shifts the focus from household payments to a larger commercial ecosystem that supports water access. This has the potential to resolve multiple of the existing challenges, like the financial struggle, professional operators and the need for volunteers to run the systems and creates at the same time a structural link between the goals of the business and the needs of the community. As the cross-subsidization solution was not found in the literature for this case it is a contribution to the literature in this context.

These findings Water Tech and similar actors to adapt their strategy before they fully launch their systems. The results point toward a phased model. This starts with building trust, supported through an initial period of grant funding and ends with a cross-subsidized revenue plan for long-term operations. The technology should prioritize local repair. Because the system was not yet in the field during this study, these findings are based on practitioner judgment and company documents. This means the contribution is about possibilities to solve the existing problems. It is not proof of this specific approach as it was not possible to observe them in practice yet.

6.2 Suggestions for Future Research

As the inductive, single-case design does not allow for generalizable findings, the most important directions for future research follow directly from the study's limitations and test the study's main contribution of the cross-subsidization model from different angles and in different contexts.

The cross-subsidization finding rests on two empirical cases, which represents a relatively thin base for a strong theoretical claim. Future research should therefore test the model across a larger number of cases with post-deployment data, with particular attention to boundary conditions. Possible questions are at what enterprise scale does cross-subsidization remain viable and what happens to community water access if the primary business fails. Another topic for future research is, how a cross subsidization model can be built up if there is no profitable business in place right now, that can cross-subsidize the water and how to approach that situation ideally.

As the OWTS was pre-commercial deployment at the time of research, a next step would be a longitudinal evaluation following actual deployment. Assessing whether the design philosophy of standardized components and local serviceability performs as intended under real operational conditions and supply chain constraints. Including the local end users and the experiences from their perspective is another point future studies should address.

Finally, since the study focused on Kenya and Zambia, comparative research across different institutional contexts, including settings with stronger or weaker state water utility involvement or peri-urban rather than purely rural communities, could help establish which elements of the framework are generalizable and which are context-specific.

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Appendix A. Interview Guide

Q1: Can you describe your role and the work you are involved in?

Q2: What is critical for a project to continue functioning over time?

Q3: What may cause a project to underperform over time?

Q4: In your view, what are the main factors that lead a company or a business owner to conclude that a technological system in a remote area is no longer sustainable to maintain?

Q5: How can responsibility and financing be structured to support sustainability of an investment?

Q6: How do incentives influence whether people maintain and repair a system over time?

Q7: We would now like you to reflect on three different ways of organizing long-term functionality. Please assume the goal is for the system to continue functioning after 10+ years.

7a) If the system is owned and operated by a group of end users, which is responsible for operation and maintenance:

What risks and opportunities do you see?

7b) If a company (manufacturer) owns the system and is responsible for operation and maintenance, financed through user fees:

What risks and opportunities do you see?

7c) If a local entrepreneur operates the system within a standardized model supported by a central actor and makes different amounts of profit based on the performance of the system:

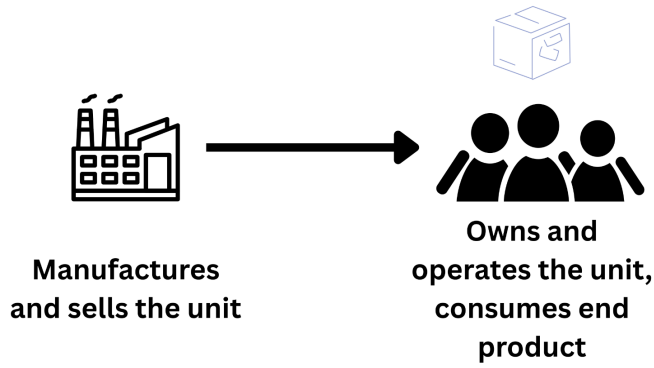
What risks and opportunities do you see?

Q8: If you were in charge of making sure a new innovative system built up in a rural area never stays broken for more than two days, how would you distribute the ownership and responsibilities to make it work?

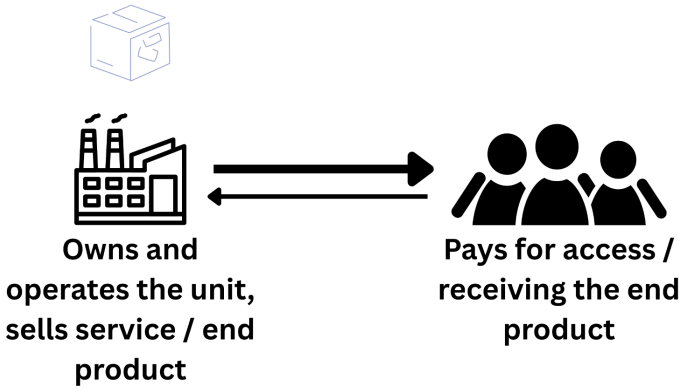
Q9: Is there anything else you would like to add?

Appendix B. Visuals for Q7 in the Interview Guide

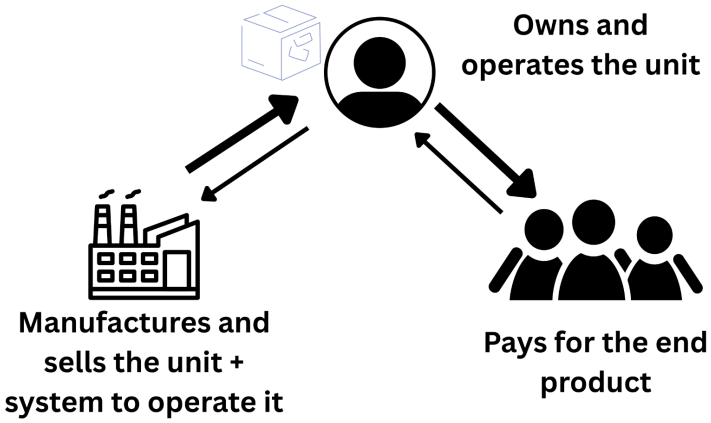
Question 7 a)



Question 7 b)



Question 7 c)



Appendix C. Documents screened

Descriptive Pseudonym	Publisher / Source
African Water Facility Strategy 2026-2030	African Development Bank: https://www.afdb.org/en/documents/african-water-facility-strategy-2026-2030
Water Tech Startup EU Project Application Zambia	Water Tech
Equator Principles	Equator Principles Limited: https://equator-principles.com/
Water Tech Startup Product Flysheet	Water Tech
GWSI Evaluation Tool	Provided by NGO-1
NGO project report	Provided by NGO-1
Social Loans - Svensk Exportkredit	Swedish Export Credit Corporation (SEK) https://www.sek.se/en/services/social-loans/
Water Tech Startup Pitch Deck	Water Tech
Sustainability Financing	Swedish Export Credit Agency (EKN) “Water is life” - EKN

Appendix D. Examples of Coding Process and Coding Table

document	1st order code	2nd order code	Theme	Quote
Local Director of Innovation and Entrepreneurship; BIA-2	Activity has to align with solving a real problem	Adapting Solution to Specific Needs of Diverse Environments	Community Involvement and Governance Structures	You could be looking at the problem statement with regards to are you solving a problem? Are you looking at a problem with solving and should you really start with that? Then, of course, part of the planning is what does it take to carry out the project, how long do you want the project to run, what are your needs, what are the things you need to fulfill the projects, and bring alignment towards the people from a point of view, from an implementation point of view, and even from a stakeholder point of view of the people who are the beneficiaries of the projects or the users of the project, bringing alignment that you can get the feedback and the results that you need
Social Enterprise Accelerator; BIA-1	Alignment between what the entrepreneur and customer want	Scaling Impact through Professionalized Social Entrepreneurship	Financial Architecture and Market Feasibility	if this is built correctly, if this is designed in a smart way, there's of course a long-term incentive to provide continuous and professional operational and maintenance services. So that sort of aligns with the interest of your end users. And this is obviously what we always strive to do, right? We have a clear synergy between what the entrepreneur wants and what the customer wants.
Water Resource Authority Research Officer; GOV-3	All stakeholders involved in a project should be held accountable if	Developing Robust Accountability and Monitoring Frameworks	Community Involvement and Governance Structures	most projects have got different stakeholders on board and there are different conversations happening at different levels that some people may not be privy to. So it all boils down to, The planners, like what are we doing and how do we ensure that everything goes

	something goes wrong as everyone has a role and part of the project			well at the end of the day. So everyone has a role to play at the end of the day.
Senior Sustainability Analyst; SSS-1	Allocate resources to support local user groups	Strengthening Local Technical Resilience	Technical Sustainability and Operational Resilience	I would allocate more resources to support the water user groups and I would also put much more resources into strengthening this network of repair persons so that they are quickly and easily accessible.
Water Tech Startup EU Project Application Zambia	Approval and import procedures can be lengthy	Harmonizing Local and International Regulatory Standards	Community Involvement and Governance Structures	Zambia requires approval from ZABS and NWASCO for the import and use of new water-treatment technologies, and procedures can be lengthy or ambiguous.
International NGO WASH Director; NGO-1	Arrogance if (white) people think they can manipulate and fix the economy of a brown person's town	Aligning Business Models with Rural Cash Flow Realities	Financial Architecture and Market Feasibility	It's not easy, and I'm not convinced that NGOs are very good at economics. There are a lot of people who think that through cash transfer programming, they can do amazing things to manipulate markets. I don't believe them, I'm sorry. You know, Nobel Prize, Nobel laureate economists don't understand the economy, but geniuses who've taken a two-week course then they can manipulate the market and economy of a small town. I'm sorry that's that's that's arrogance and I'm sorry a little bit of racism when a white person's telling me that they can they can fix the economy of a brown person's town.

Note: The full coding table will be made available on request.